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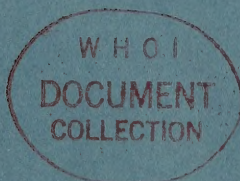
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A NONLINEAR MATHEMATICAL MODEL OF MOTIONS OF A PLANING BOAT IN REGULAR WAVES

by

Ernest E. Zarnick



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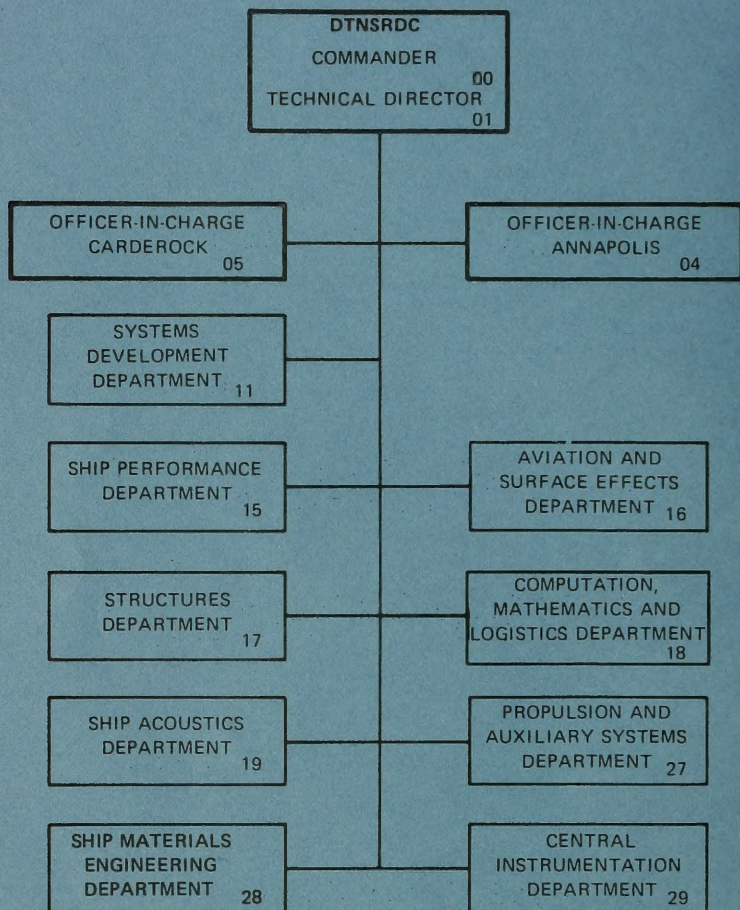
SHIP PERFORMANCE DEPARTMENT
RESEARCH AND DEVELOPMENT REPORT

March 1978

DTNSRDC-78/032

GC
D3
no. 78/032

MAJOR DTNSRDC ORGANIZATIONAL COMPONENTS



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0 2662600 1060 0 0301 0032532 0

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SECURITY CLASSIFICATION OF THIS PAGE (When Data Entered)

REPORT DOCUMENTATION PAGE		READ INSTRUCTIONS BEFORE COMPLETING FORM
1. REPORT NUMBER DTNSRDC-78/032	2. GOVT ACCESSION NO.	3. RECIPIENT'S CATALOG NUMBER
4. TITLE (and Subtitle) A NONLINEAR MATHEMATICAL MODEL OF MOTIONS OF A PLANING BOAT IN REGULAR WAVES		5. TYPE OF REPORT & PERIOD COVERED
		6. PERFORMING ORG. REPORT NUMBER
7. AUTHOR(s) Ernest E. Zarnick		8. CONTRACT OR GRANT NUMBER(s)
9. PERFORMING ORGANIZATION NAME AND ADDRESS David W. Taylor Naval Ship Research and Development Center Bethesda, Maryland 20084		10. PROGRAM ELEMENT, PROJECT, TASK AREA & WORK UNIT NUMBERS ZF 43 421001 Work Unit 1-1500-100
11. CONTROLLING OFFICE NAME AND ADDRESS Naval Sea Systems Command (SEA 035) Washington, D.C. 20362		12. REPORT DATE March 1978
		13. NUMBER OF PAGES 86
14. MONITORING AGENCY NAME & ADDRESS (if different from Controlling Office)		15. SECURITY CLASS. (of this report) UNCLASSIFIED
		15a. DECLASSIFICATION/DOWNGRADING SCHEDULE
16. DISTRIBUTION STATEMENT (of this Report) APPROVED FOR PUBLIC RELEASE: DISTRIBUTION UNLIMITED		
17. DISTRIBUTION STATEMENT (of the abstract entered in Block 20, if different from Report)		
18. SUPPLEMENTARY NOTES		
19. KEY WORDS (Continue on reverse side if necessary and identify by block number) Planing Boat Motions Hydrodynamic Impact Small Boat Worthiness Nonlinear Ship Motions in Waves		
20. ABSTRACT (Continue on reverse side if necessary and identify by block number) A nonlinear mathematical model has been formulated of a craft having a constant deadrise angle, planing in regular waves, using a modified low-aspect-ratio or strip theory. It was assumed that the wavelengths would be large in comparison to the craft length and that the wave slopes would be small. The coefficients in the equations of motion were determined by a combination of theoretical and empirical relationships. A simplified version for the case of a craft or model being towed at constant speed was programmed for computations on a digital computer, and the results were compared with existing experimental data.		

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Comparison of computed pitch and heave motions and phase angles with corresponding experimental data was remarkably good. Comparison of bow and center of gravity vertical accelerations was fair to good.

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NOTATION

A	Mass matrix
A_R	Section area
a	Correction factor for buoyancy force
b	Half-beam of craft
$C_{D,c}$	Crossflow drag coefficient
C_Δ	Load coefficient $\Delta/pg(2b)^3$
C_λ	Wavelength coefficient $L/\lambda [C_\Delta/(L/2b)^2]^{1/3}$
D	Friction drag force
F_x	Total hydrodynamic force in x direction
F_z	Total hydrodynamic force in z direction
F_θ	Total hydrodynamic moment about pitch axis
f	Two-dimensional hydrodynamic force
g	Acceleration of gravity
H	Wave height, crest to trough
h	Vertical submergence of point below free surface
h_z	Double amplitude of heave
I	Pitch moment of inertia
I_a	Added pitch, moment of inertia
k	Wave number
k_a	Two-dimensional added-mass coefficient
L	Hull length
LCG	Longitudinal center of gravity, percent of L
M	Mass of craft
M_a	Added mass of craft

m_a	Sectional (two-dimensional) added mass
N	Hydrodynamic force normal to baseline
r	Wave elevation $r = r_o \cos(kx + \omega t)$
r_o	Wave amplitude
U	Relative fluid velocity parallel to baseline
V	Relative fluid velocity normal to baseline
$V/\sqrt{L}$	Speed-to-length ratio in knots/ft ^{1/2}
W	Weight of craft
w_z	Vertical component of wave orbital velocity
$\dot{w}_z$	Vertical component of wave orbital acceleration
x	Fixed horizontal coordinate
$\bar{x}$	Vector of state variables
$\dot{x}_{CG}$	Surge velocity
$\ddot{x}_{CG}$	Surge acceleration
x_{CG}	Surge displacement
z	Fixed vertical coordinate
$\dot{z}_{CG}$	Heave velocity
$\ddot{z}_{CG}$	Heave acceleration
z_{CG}	Heave displacement
β	Deadrise angle
Δ	Hull displacement W
ζ	Body coordinate normal to baseline
λ	Wavelength
θ	Pitch angle
$\dot{\theta}$	Pitch angular velocity

ABSTRACT

A nonlinear mathematical model has been formulated of a craft having a constant deadrise angle, planing in regular waves, using a modified low-aspect-ratio or strip theory. It was assumed that the wavelengths would be large in comparison to the craft length and that the wave slopes would be small. The coefficients in the equations of motion were determined by a combination of theoretical and empirical relationships. A simplified version for the case of a craft or model being towed at constant speed was programed for computations on a digital computer, and the results were compared with existing experimental data. Comparison of computed pitch and heave motions and phase angles with corresponding experimental data was remarkably good. Comparison of bow and center of gravity vertical accelerations was fair to good.

ADMINISTRATIVE INFORMATION

This investigation was authorized by the Naval Sea Systems Command with initial funding under Task Area SR-023-0101 and completion under Task Area ZF-43-421001.

INTRODUCTION

Computer programs for estimating the motions of displacement ships in waves for all headings and speeds have been in existence for some time. Comparable computational schemes for planing craft do not exist except in limited and restricted cases. A program for planing craft would be quite useful to the small craft designer, providing a means for systematically exploring the effects of numerous design variations on performance of the craft in waves. With minor modification, the program could also be used to examine the merits of a hybrid craft design, e.g., a combination of planing craft and hydrofoil.

Predicting the motions of a planing craft in wave's is by no means a simple problem. The analytical description of a high-speed craft, planing in waves, involves several different types of flow phenomena, including planing; hydrodynamic impact, and, to a lesser extent, surface wave generation and hydrostatics. Also, the mathematics tend to become nonlinear rapidly as the motion increases or, like the real craft, can in some instances exhibit large instabilities such as porpoising.

Development of a computer program that would take into account all of the previously described factors and would be applicable for a wide range of speed and wave conditions requires a careful and systematic study in several stages with appropriate verification at each stage. To lay the foundation for such a general program, a simpler problem has been

formulated in this report with potential for expansion and generalization to the more complicated case. The simpler problem is that of a V-shaped prismatic body with hard chines and constant deadrise planing at high speed in regular head waves.

The mathematical formulation is analogous to low-aspect-ratio wing theory with provisions for including hydrodynamic impact loads, essentially a strip theory. Surface wave generation and forces associated with unsteady circulatory flow are neglected, and the flow is treated as quasi-steady. The mathematical formulation is an empirical synthesis of several theoretically derived flows describing the overall craft hydrodynamics. Wave input is restricted to monochromatic linear deepwater waves with moderate wavelengths and low wave slopes.

MATHEMATICAL FORMULATION

GENERAL

Consider a fixed coordinate system (x, z) (Figure 1) with x axis in the undisturbed free surface, pointing in the direction of craft travel, and the z axis, pointing downward. If the motions of the craft are restricted to pitch θ , heave z_{CG} , and surge x_{CG} , the equation of motions can be written as

$$\begin{aligned} M\ddot{x}_{CG} &= T_x - N \sin \theta - D \cos \theta \\ M\ddot{z}_{CG} &= T_z - N \cos \theta + D \sin \theta + W \\ I\ddot{\theta} &= Nx_c + Dx_d + Tx_p \end{aligned} \quad (1)$$

where M is mass of craft

I is pitch moment of inertia of craft

N is hydrodynamic normal force

D is friction drag

W is weight of craft

T_x is thrust component in x direction

T_z is thrust component in z direction

x_c is distance from center of gravity (CG) to center of pressure for normal force

x_d is distance from CG to center of action for friction drag force

x_p is moment arm of thrust about CG.

Equation (1) is exact; however, defining the hydrodynamic forces and moments in waves can be extremely difficult.

A high-speed craft moving in waves may transit through several regimes that have different hydrodynamic flow characteristics. For example, as the craft moves away from the crest of wave, the flow may be characterized by unsteady-state planing until the craft collides with the oncoming wave crest and enters another regime in which impact forces are important. After the impact, the craft may enter still another regime in which it is planing but in which buoyancy forces are rather significant.

The most promising approach to a method that would incorporate all three types of flow conditions into a general formulation would seem to be a modified strip theory. The mathematical justification for this approach is not rigorous; however, there is sufficient precedent to expect promising results. For example, impact loads on landing seaplanes can be estimated reasonably well using a strip theory incorporating the Wagner two-dimensional (2-D), expanding-wedge theory,¹ and Chuang² has provided a strip method for determining loads on an impacting prismatic form that agrees extremely well with experimental results.

More recently, Martin³ has developed a linear strip theory for estimating motions of a planing craft at high speed, which shows good agreement with experimental results. A nonlinear model of the equations of motion would be expected to provide, in addition to the motions, reasonable estimates of the vertical accelerations which are an important consideration in designing a planing craft.

TWO-DIMENSIONAL HYDRODYNAMIC FORCE

Implicit with any strip method is the need to define the 2-D hydrodynamic force acting on an arbitrary cross section of the body. The 2-D flow problem is not simple; however, it lends itself to an empirical approach, using a combination of techniques used in hydrodynamic impact and low-aspect-ratio theories.

The typical cross section of a hard-chine, V-shaped prismatic body such as that being considered here is shown in Figure 2. Figure 2 actually illustrates two different idealized-flow conditions, assumed to represent the crossflow during unsteady planing, depending upon whether the flow separates from the chine (Figure 2a) or not (Figure 2b). Nonwetted-chine flow conditions are typical of the sections near the leading edge of the wetted length of the craft. Wetted-chine flow conditions are more typical of sections near the stern, except possibly in the most extreme motion and wave conditions. Some sections between leading edge and stern may alternate between flow conditions as the wetted length changes with the motions.

*A complete listing of references is given on page 33.

The normal hydrodynamic force per unit length f , acting at a section, is treated as quasi-steady and is assumed to contain components proportional to the rate of change of momentum and the velocity squared (drag term), i.e.

$$f = - \left\{ \frac{D}{Dt} (m_a V) + C_{D,c} \rho b V^2 \right\} \quad (2)$$

where V is the velocity in plane of the cross section normal to the baseline

m_a is the added mass associated with the section form

$C_{D,c}$ is the crossflow drag coefficient

ρ is the density of the fluid

b is the half beam.

For sections near the leading edge of the wetted length with nonwetted chine, the added mass is assumed to be defined in the same manner as during an impact which for a V-shaped wedge is given by

$$m_a = k_a \pi/2 \rho b^2 \quad (3)$$

where k_a is an added-mass coefficient that may also include a correction for water pileup – k_a is assumed to be 1.0 without pileup correction.

The rate of change of momentum of the fluid at a section is given by

$$\frac{D}{Dt} (m_a V) = m_a \dot{V} + V \dot{m}_a - \frac{\partial}{\partial \xi} (m_a V) \frac{d\xi}{dt} \quad (4)$$

where ξ is the body coordinate parallel to the baseline; see Figure 1. The last term on the right-hand side of Equation (4) takes into account the variation of the section added mass along the hull. This contribution can be visualized by considering the 2-D flow plane as a substantive surface moving past the body with velocity $U = -d\xi/dt$ tangent to the baseline. As the surface moves past the body, the section geometry in the moving surface may change with a resultant change in added mass. This term exists even in steady-state conditions and is the lift-producing factor in low-aspect-ratio theory.

The added mass of a section with fully wetted chines has not been developed to the same extent as the V wedge. In steady-state planing problems such as those of Shuford,⁴

the crossflow is treated as a Helmholtz-type flow in which the Bobyleff results are used for estimating drag coefficients. Helmholtz flows are applicable only to steady-state conditions; so, it is assumed that the added mass for the fully wetted chine flow can be determined from Equation (3) using the value of the half-beam at the chine. In using the Shuford approach, it is assumed that the crossflow drag coefficient for a V-section is equal to the drag of a flat plate ($C_{D,c} = 1.0$) corrected by the Bobyleff flow coefficient approximated by $\cos \beta$, i.e.

$$C_{D,c} = 1.0 \cos \beta \quad (5)$$

The Bobyleff flow coefficient is the theoretical ratio of the pressure on a V-section to that experienced by a flat plate for a Helmholtz-type flow.

The same approximation is used for estimating the drag coefficient for nonwetted chine sections, using the instantaneous value of the half-beam at the free surface.

An additional force acting on the body is the buoyancy force f_B . This force is assumed herein to act in the vertical direction and to be equal to the equivalent static buoyancy force multiplied by a correction factor, i.e.

$$f_B = -a\rho g(A) \quad (6)$$

where A is the cross-sectional area of the section, and a is a correction factor.

The full amount of the static buoyancy is not realized because at planing speeds the water separates from the transom and chines, reducing the pressure at these locations to atmospheric or less than the equivalent hydrostatic pressure. A greater reduction is realized in the buoyancy moment because of the corresponding shift in the center of pressure. Shuford⁴ in his work on steady-state planing recommended a factor of one-half to obtain the correct buoyancy force. In the following computations, the buoyancy force was corrected by a factor of one-half, i.e., $a = 1/2$. The buoyancy moment, computed as the static buoyancy force multiplied by its corresponding moment arm, was corrected by an additional factor of one-half to obtain the proper mean-trim angles.

Equation (2) is a synthesis of several idealized flow conditions combined in an empirical manner. In all of these flows, it is assumed that the net relative movement of the fluid past the body is in an upward direction. This condition may not always be met in the case of unsteady planing in waves. Closer scrutiny will be required to determine what limitations will be imposed upon the problem as formulated and/or what modifications will be required to improve the formulation.

TOTAL HYDRODYNAMIC FORCE AND MOMENT

The total normal hydrodynamic force acting on the body is obtained by integrating the stripwise, 2-D, hydrodynamic force given by Equations (2) and (6) over the wetted length ℓ of the body. A body coordinate system (ξ, ζ) with its origin at CG and the ξ axis pointing forward parallel to the baseline of the body is defined in Figure 1 to facilitate this integration. The hydrodynamic force acting in the vertical or z direction of the fixed integral coordinate system is given by

$$\begin{aligned}
 -N \cos \theta &= F_z(t) = \int_{\ell} f \cos \theta \, d\xi + \int_{\ell} f_B \, d\xi \\
 &= - \left[\int_{\ell} \left\{ m_a(\xi, t) \dot{V}(\xi, t) + \dot{m}_a(\xi, t) V(\xi, t) \right. \right. \\
 &\quad - U(\xi, t) \frac{\partial}{\partial \xi} [m_a(\xi, t) V(\xi, t)] \\
 &\quad + C_{D,c}(\xi, t) \rho b(\xi, t) V^2(\xi, t) \left. \right\} \cos \theta \, d\xi \\
 &\quad \left. + a \rho g A \, d\xi \right] \quad (7)
 \end{aligned}$$

where the integration is taken over the instantaneous wetted length. Similarly the force F_x acting in the horizontal or x direction is given by

$$\begin{aligned}
 F_x &= \int_{\ell} f \sin \theta \, d\xi \\
 &= - \int_{\ell} \left\{ m_a(\xi, t) \dot{V}(\xi, t) + \dot{m}_a(\xi, t) V(\xi, t) \right. \\
 &\quad - U(\xi, t) \frac{\partial}{\partial \xi} [m_a(\xi, t) V(\xi, t)] \\
 &\quad \left. + C_{D,c}(\xi, t) \rho b(\xi, t) V^2(\xi, t) \right\} \sin \theta \, d\xi \quad (8)
 \end{aligned}$$

Wave forces are obtained by neglecting diffraction and assuming that the wave excitation is caused both by the geometrical properties of the wave, altering the wetted length and draft of the craft, and by the vertical component of the wave orbital velocity at the surface w_z , altering the normal velocity V . The horizontal component of orbital velocity is neglected,

since it is assumed small in comparison with the forward speed $\dot{x}_{CG}$. The velocities U and V may then be written as

$$\begin{aligned} U &= \dot{x}_{CG} \cos \theta - (\dot{z}_{CG} - w_z) \sin \theta \\ V &= \dot{x}_{CG} \sin \theta - \dot{\theta} \xi + (\dot{z}_{CG} - w_z) \cos \theta \end{aligned} \quad (9)$$

The depth of submergence h of the body at any point P(ξ, ζ) may be determined by

$$h = z_{CG} - \xi \sin \theta + \zeta \cos \theta - r \quad (10)$$

where r is the instantaneous value of the wave elevation directly above the point.

For regular head waves the wave elevation for a linear deepwater wave is

$$r = r_0 \cos k(x+ct) \quad (11)$$

where r_0 is the wave amplitude

k is the wave number

c is the wave celerity.

At point P(ξ, ζ)

$$x = x_{CG} + \xi \cos \theta + \zeta \sin \theta \quad (12)$$

where $x_{CG} = \int_{\ell} \dot{x}_{CG} dt$

The hydrodynamic moment F_θ about CG is obtained in a similar manner by integrating over the wetted length the product of the normal force per unit length and the corresponding moment arm.

$$\begin{aligned}
F_\theta &= - \int_{\ell} f(\xi, t) \xi d\xi - \int_{\ell} f_b \cos \theta \xi d\xi \\
&= \int_{\ell} \left\{ m_a(\xi, t) \dot{V}(\xi, t) + \dot{m}_a(\xi, t) V(\xi, t) \right. \\
&\quad \left. - U(\xi, t) \frac{\partial}{\partial \xi} (m_a(\xi, t) V(\xi, t)) + C_{D,c}(\xi, t) \rho b(\xi, t) V^2(\xi, t) \right. \\
&\quad \left. + a \rho g A \cos \theta \right\} \xi d\xi
\end{aligned} \tag{13}$$

EQUATIONS OF MOTION, GENERAL

Integrating the first term in Equations (7), (8), and (13) provides hydrodynamic forces and moments proportional to acceleration of the motion. These can be combined with the inertial terms of the rigid body to give the following equation of motion

$$\begin{aligned}
&(M + M_a \sin^2 \theta) \ddot{x}_{CG} + (M_a \sin \theta \cos \theta) \ddot{z}_{CG} - (Q_a \sin \theta) \ddot{\theta} \\
&= T_x + F'_x - D \cos \theta \\
&(M_a \sin \theta \cos \theta) \ddot{x}_{CG} + (M + M_a \cos^2 \theta) \ddot{z}_{CG} - (Q_a \cos \theta) \ddot{\theta} \\
&= T_z + F'_z + D \sin \theta + W \\
&-(Q_a \sin \theta) \ddot{x}_{CG} - (Q_a \cos \theta) \ddot{z}_{CG} + (I + I_a) \ddot{\theta} \\
&= F'_\theta - D x_d + T x_p
\end{aligned} \tag{14}$$

where $M_a(t) = \int_{\ell} m_a(\xi, t) d\xi$

$$Q_a(t) = \int_{\ell} m_a(\xi, t) \xi d\xi$$

$$I_a(t) = \int_{\ell} m_a(\xi, t) \xi^2 d\xi$$

$$F'_x = F_x - \left\{ -(M_a \sin^2 \theta) \ddot{x}_{CG} - (M_a \sin \theta \cos \theta) \ddot{z}_{CG} + (Q_a \sin \theta) \ddot{\theta} \right\}$$

$$F'_z = F_z - \left\{ \text{appropriate acceleration terms} \right\}$$

$$F'_\theta = F_\theta - \left\{ \text{appropriate acceleration terms} \right\}.$$

A detailed evaluation of the integral expressions for the hydrodynamic forces and moments is provided in Appendix A.

The solution to Equation (14) is cumbersome; however, it can be accomplished using standard numerical techniques. Introducing the state vector $[x_1, x_2, x_3, x_4, x_5, x_6]$

$$\text{where } x_1 = \dot{y}_{CG}$$

$$x_2 = \dot{z}_{CG}$$

$$x_3 = \dot{\theta}$$

$$x_4 = x_{CG}$$

$$x_5 = z_{CG}$$

$$x_6 = \theta$$

Equation (14) can be rewritten, using matrix algebra, as

$$A \vec{\dot{x}} = \vec{g} \quad (15)$$

so that

$$\vec{\dot{x}} = A^{-1} \vec{g} \quad (16)$$

where A^{-1} is inverse of the inertial matrix A . Equation (16) is now in a form that lends itself to integration by using a numerical method such as the Runge-Kutta-Merson integration routine.

EQUATIONS OF MOTION, SIMPLIFIED FOR CONSTANT SPEED

Assuming that the perturbation velocities in the forward direction are small in comparison to the speed of the craft, the equations of motion may be further simplified by neglecting the perturbations and setting the forward velocity equal to a constant, i.e.

$$\dot{x}_{CG} = \text{CONSTANT}$$

If it is also assumed that the thrust and drag forces are small in comparison to the hydrodynamic forces and that they are acting through the center of gravity, the equations of motion may be written as

$$\begin{aligned}\ddot{x}_{CG} &= 0 \\ (M + M_a \cos^2 \theta) \ddot{z}_{CG} - (Q_a \cos \theta) \ddot{\theta} &= F'_z + W \\ -(Q_a \cos \theta) \ddot{z}_{CG} + (I + I_a) \ddot{\theta} &= F'_\theta\end{aligned}$$

These equations also represent the case of the craft (model) being towed through CG at CONSTANT speed. Based upon the previously described equations of motion, a computer program has been written in FORTRAN language to compute the motions of a prismatic body, planing in regular head waves at high speed. A listing of the program along with the appropriate flow chart is presented in Appendix B. The listing contains reference to thrust and drag terms; however, they have no significance, except to provide a starting point for possible updating of the program to include these terms in the future.

COMPARISON OF COMPUTED RESULTS WITH EXPERIMENTS

Computations of pitch and heave motions and heave and bow accelerations were made, using the computer program for comparison with the experimental results of Fridsma.⁵ Fridsma tested a series of constant-deadrise models of various lengths in regular waves to define the effects of deadrise, trim, loading, speed, length-to-beam ratio and wave proportions on the added resistance, heave and pitch motions, and impact accelerations at the bow and center of gravity. Figure 3 shows the lines of the prismatic models. The models were towed at CG with a system that permitted freedom in surge. The computer program simulates the model being towed at constant speed with CG at the baseline.

Table 1 presents some characteristics of the model and experimental conditions for which comparisons were made. Most of the comparisons have been made at a speed-to-length ratio $V/\sqrt{L}$ of 6.0 where the mathematical model is expected to be most representative. A limited comparison has also been made at $V/\sqrt{L} = 4.0$; however, no comparison has been made at $V/\sqrt{L} = 2.0$. At this speed, the model (or craft) operates in the displacement mode for which the mathematical formulation is not valid.

The average computer run corresponded to 10-second, real-time, model scale; however, only the last 2 seconds were considered free of transient effects. An example of the computer time histories of pitch and heave motions is shown in Figure 4. Although the motions are periodic, they are not perfectly sinusoidal; consequently, in determining phase relationship, the peak, positive-pitch value (bow up) and the peak, negative-heave value (maximum upward position of CG) were used as reference points. There was a difference when the opposite peaks were used.

**TABLE 1 – MODEL CHARACTERISTICS AND WAVE
CONDITIONS FOR COMPUTATIONS**

(Model Length = 114.3 cm (3.75 ft); L/b = 5; $C_{\Delta} = 0.608$)

CONFIGURATIONS							
SYMBOL	β deg	LCG percent L	Radius of Gyration percent L	$v/\sqrt{L}$			
A	20	59.0	25.1	4.0			
B	20	62.0	25.5	6.0			
J	10	68.0	26.2	6.0			
M	30	60.5	24.8	6.0			
WAVE CONDITIONS FOR CONFIGURATION --							
A		B		J		M	
$\frac{H}{b}$	$\frac{\lambda}{L}$	$\frac{H}{b}$	$\frac{\lambda}{L}$	$\frac{H}{b}$	$\frac{\lambda}{L}$	$\frac{H}{b}$	$\frac{\lambda}{L}$
0.111	1.0	0.111	1.0	0.111	1.0	0.111	1.0
0.111	1.5	0.111	1.5	0.111	1.5	0.111	1.5
0.111	2.0	0.111	2.0	0.111	2.0	0.111	2.0
0.111	3.0	0.111	3.0	0.111	3.0	0.111	3.0
0.111	4.0	0.111	4.0	0.111	4.0	0.111	4.0
0.111	6.0	0.222	6.0	0.111	6.0	0.111	6.0
		0.334	4.0				
		0.111	6.0				

Corresponding time histories of bow and CG accelerations are shown in Figure 5. The bow acceleration was computed at Station 0. As can be seen in these plots, the impact accelerations ranged in magnitude from cycle to cycle. The maximum impact (or negative value) acceleration computed during the final 2 seconds of run was used in the comparisons with experimental values. In some instances, particularly near resonance, the maximum impact acceleration was more than twice the average impact value.

Figure 6 shows a comparison of variation of computed and experimental pitch and heave motion with wave height for the 20-degree deadrise model in a 15-foot wavelength and for a speed-to-length ratio of 6.0. Figure 7 shows the corresponding impact acceleration at the bow and CG. The computed results closely follow the experimental data, except for CG acceleration at the extreme wave height condition, where the computed value is apparently much lower. Experimental data show that the model was leaving the water at this wave-height condition. The computer model did not leave the water but came very close;

see Figure 8. Figure 8 is a trajectory of the computer model relative to the wave for a selected cycle of motion. The computer model behaves very much as expected. On the left-hand side of the figure, the craft is planing down the crest of the wave and, as it approaches the wave trough, comes very close to leaving the water before slamming and submerging itself deeply into the front of the oncoming wave crest.

Figures 9 through 14 show comparisons of the computed and experimental pitch and heave motions at $V/\sqrt{L} = 6.0$ through a range of wavelengths and at a constant wave height of 2.54 centimeters (1 inch) for deadrise models with 10, 20, and 30 degrees. The data have been plotted with respect to the coefficient C_λ , defined by Fridsma as $L/\lambda [C_\Delta/(L/2b)^2]^{1/3}$. Note that in our notation, b is the half-beam.

Comparisons of heave and pitch for the 10-degree deadrise model shown in Figures 9 and 10, respectively, show excellent results. The computer model accurately predicts the secondary peaks in the pitch and heave responses at $C_\lambda = 0.19$. At this condition, the physical experimental model rebounds so as to fly over alternate waves. The computer model oscillates at half the wave-encounter frequency and comes close to leaving the water at alternate encounters with the wave. It does not quite leave the water to fly over alternate wave crests; nonetheless, it is a good representation of the actual motion.

The heave and pitch comparison for the 20-degree deadrise model at $V/\sqrt{L} = 6.0$ is also excellent as can be seen in Figures 11 and 12, respectively. No experimental phase data for the condition were reported for C_λ greater than 0.072; however, extrapolated results (not shown) are in line with the computed results. The pitch and heave results shown in Figures 13 and 14 for the 30-degree deadrise model are good; however, responses at $C_\lambda = 0.048$ and $C_\lambda = 0.072$ are higher than the experimental results.

For practical considerations a computational scheme for planing boat motions should be valid for a range from approximately $V/\sqrt{L} = 4.0$ to $V/\sqrt{L} = 6.0$. Computations of the motions were made for $V/\sqrt{L} = 4.0$ for the 20-degree deadrise model; see Figures 15 and 16. Again the comparison of the computed heave and pitch response with experimental results is excellent.

Comparisons of the computed and experimental impact accelerations (or largest negative values) are presented in Figures 17 through 20. Figures 17 and 18 show bow and CG accelerations for the 10-degree deadrise model; Figure 19 shows similar results for the 20-degree deadrise model; Figure 20 shows the results for the 30-degree deadrise model. In all cases, the comparison appears to be fair to good. In the shorter wavelengths, $\lambda/L = 1.0$ and $\lambda/L = 1.5$, the computed accelerations are higher than the corresponding experimental values. This is most pronounced for the 10-degree deadrise angle model.

CONCLUSIONS AND RECOMMENDATIONS

A mathematical model of a craft having a constant deadrise angle, planing in regular waves, has been formulated using a modified low-aspect-ratio or strip theory. It was assumed that the wavelengths were long in comparison to the craft length and that the wave slopes were small. The coefficients in the equations of motion were determined by a combination of theoretical and empirical relationships.

A simplified version for the case of a craft or model being towed at constant speed was programmed for computations on a digital computer, and the results were compared with existing experimental data.

The comparison of the computed pitch and heave motions and phase angles with the corresponding experimental data gave remarkably satisfying results. Comparison of the bow and CG accelerations was fair to good.

In summary, the previously described mathematical model appears to be a valid representation of a planing craft in waves for the specific craft geometry and wave conditions considered.

To make the computer program more valuable to the designer the following additional work is recommended:

1. Improve estimates of hydrodynamic coefficients to obtain better acceleration data and to include more complicated ship geometry.
2. Determine added resistance in waves.
3. Include freedom to surge and to add components of propulsion.
4. Extend to the case of irregular waves.

ACKNOWLEDGMENTS

Acknowledgment is given to Dr. Joseph Whalen and Ms. Sue Fowler of Operations Research, Inc., who translated the equations of motion into an operational computer program.

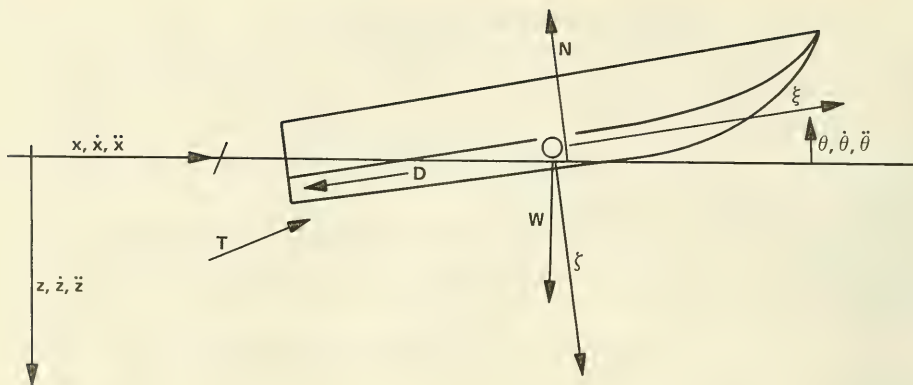


Figure 1 – Coordinate System

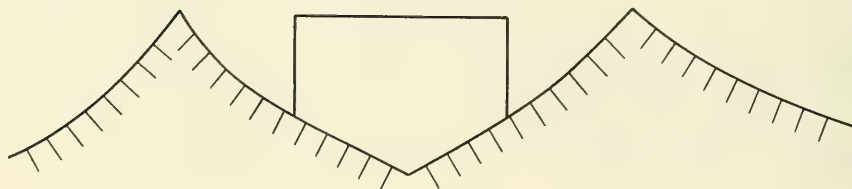


Figure 2a – Flow Separation from Chine

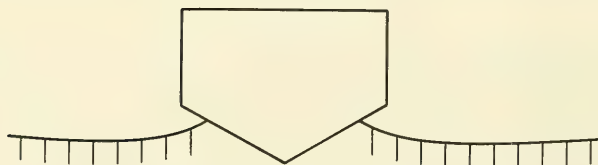


Figure 2b – Nonwetted Chine

Figure 2 – Types of Two-Dimensional Flow

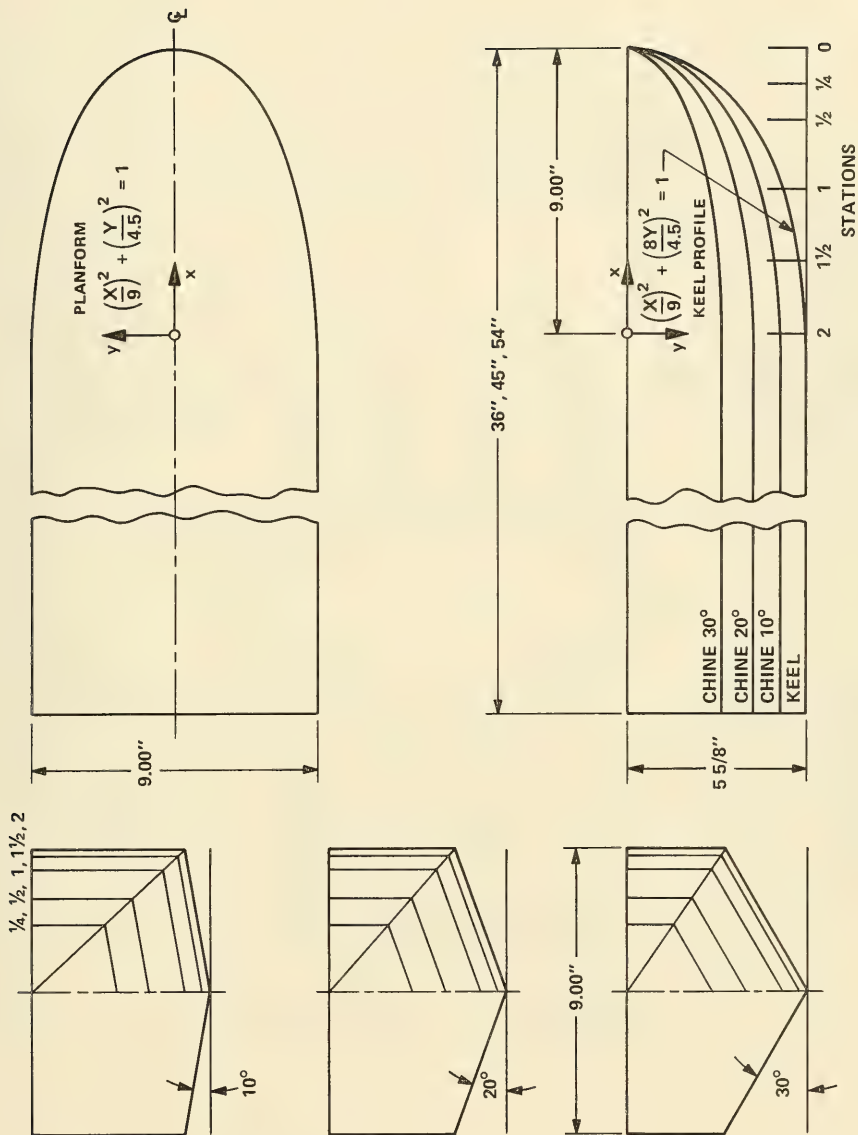


Figure 3 – Lines of Prismatic Models
(From Reference 5)

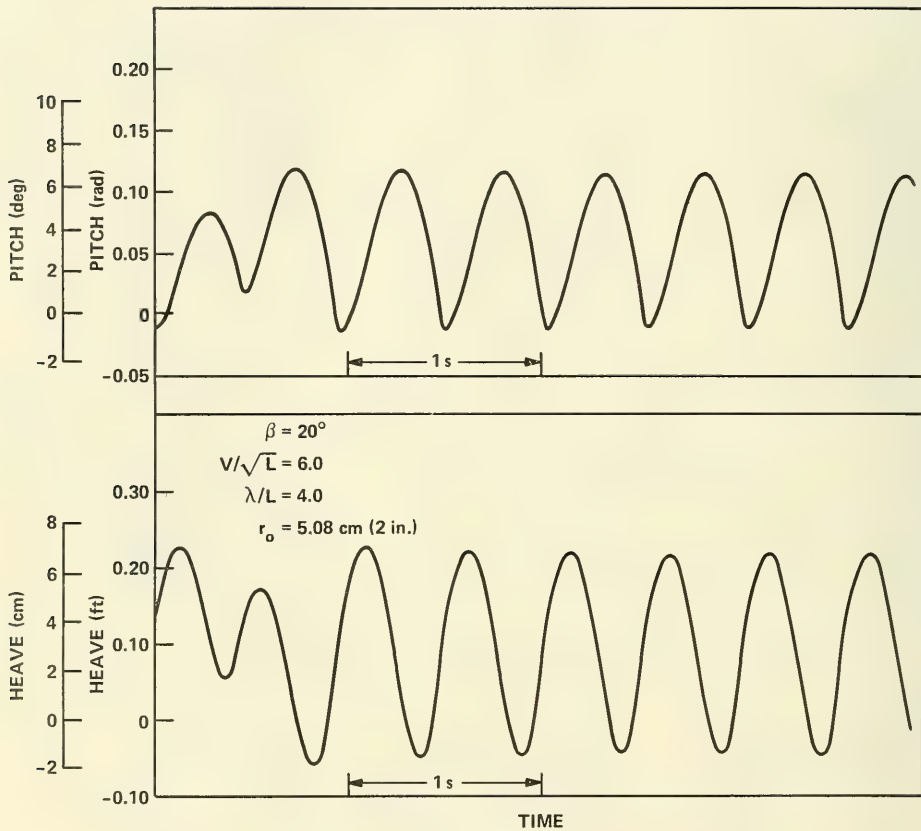


Figure 4 – Sample Time Histories of Computed Pitch and Heave Motions

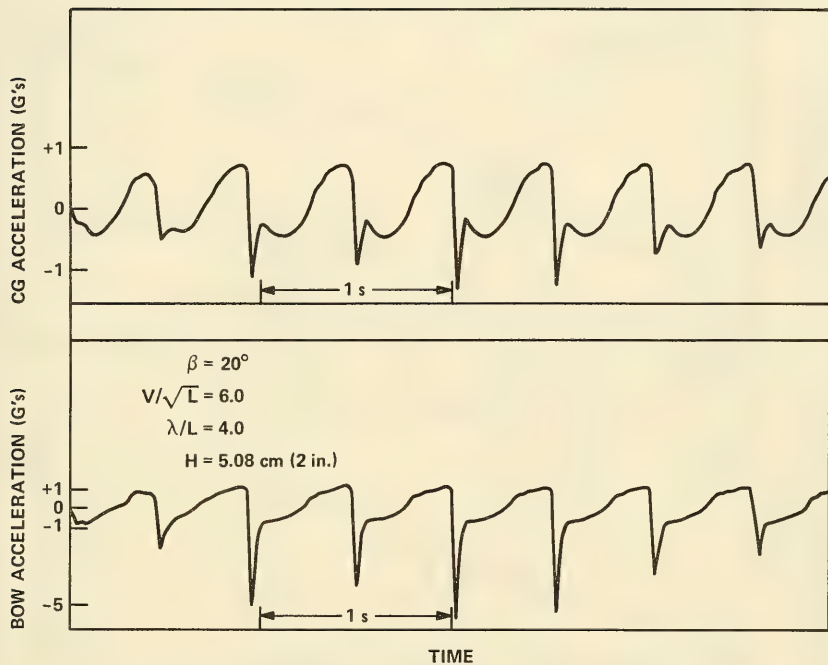


Figure 5 — Sample Time Histories of Computed Accelerations of Bow and Center of Gravity

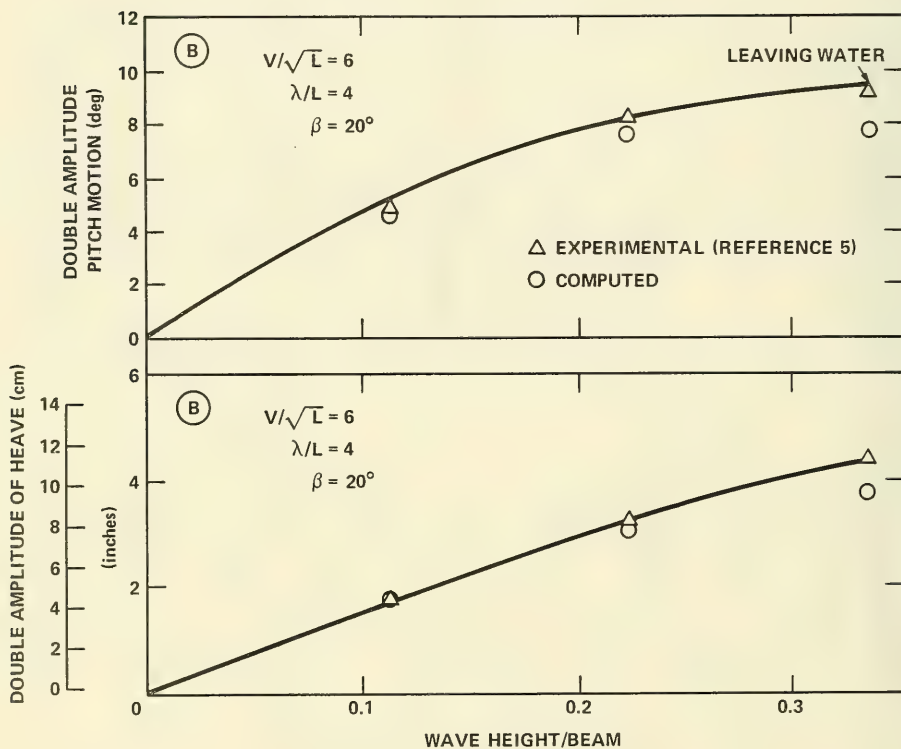


Figure 6 – Variation of Pitch and Heave with Wave Height

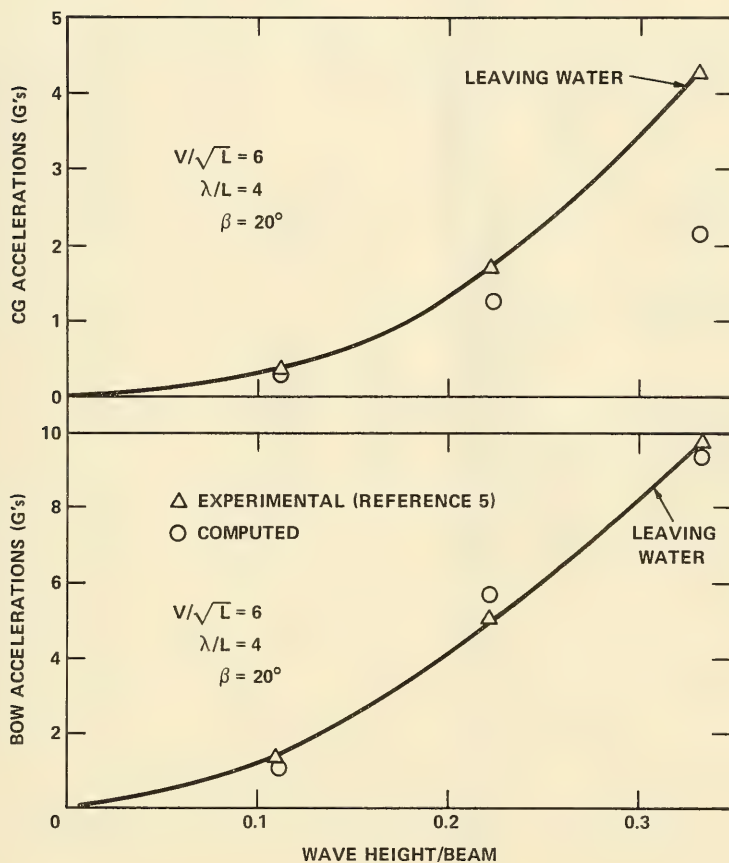


Figure 7 - Variation of Acceleration of Bow and Center of Gravity with Wave Height

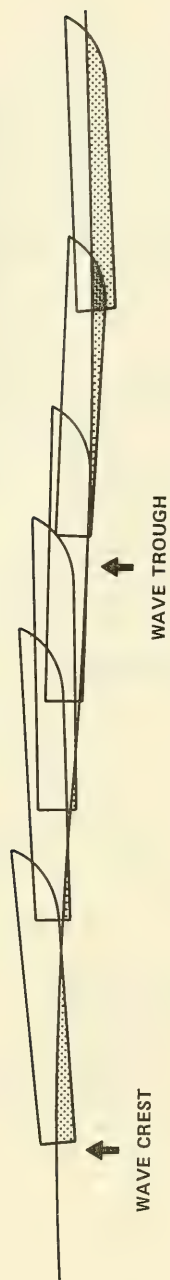


Figure 8 – Trajectory of Computer Model Relative to Wave

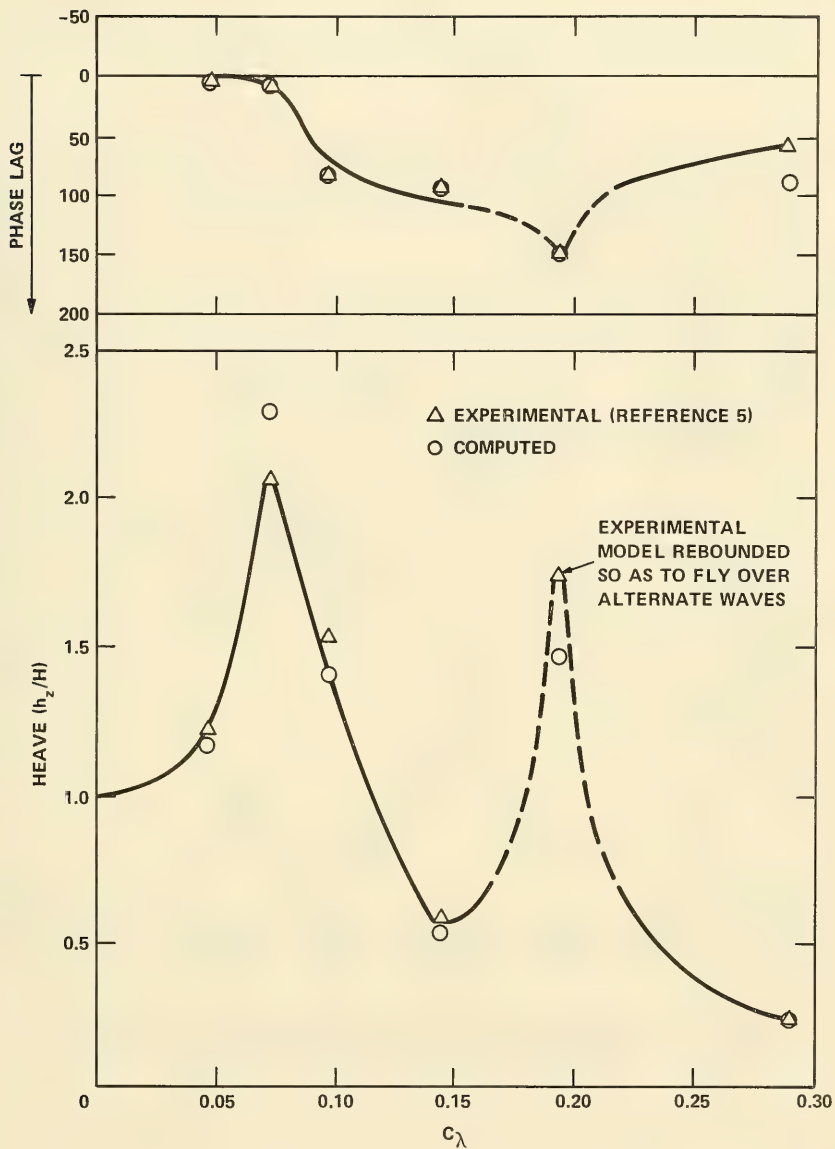


Figure 9 – Heave Response for 10-Degree Deadrise Model at $V/\sqrt{L} = 6.0$

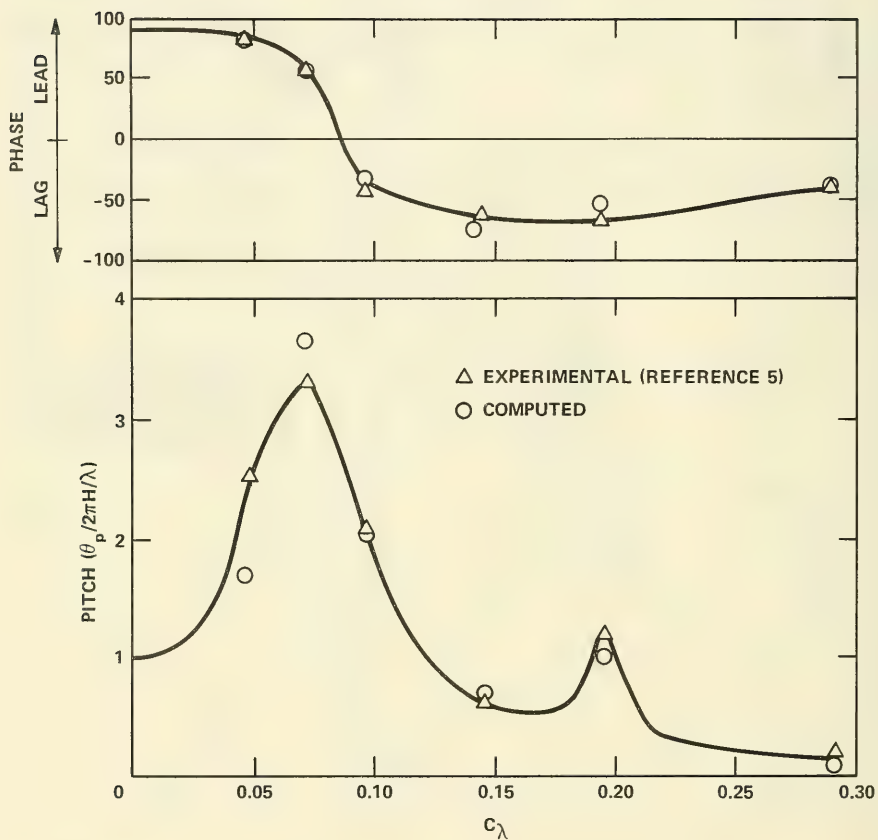


Figure 10 – Pitch Response for 10-Degree Deadrise Model at $V/\sqrt{L} = 6.0$

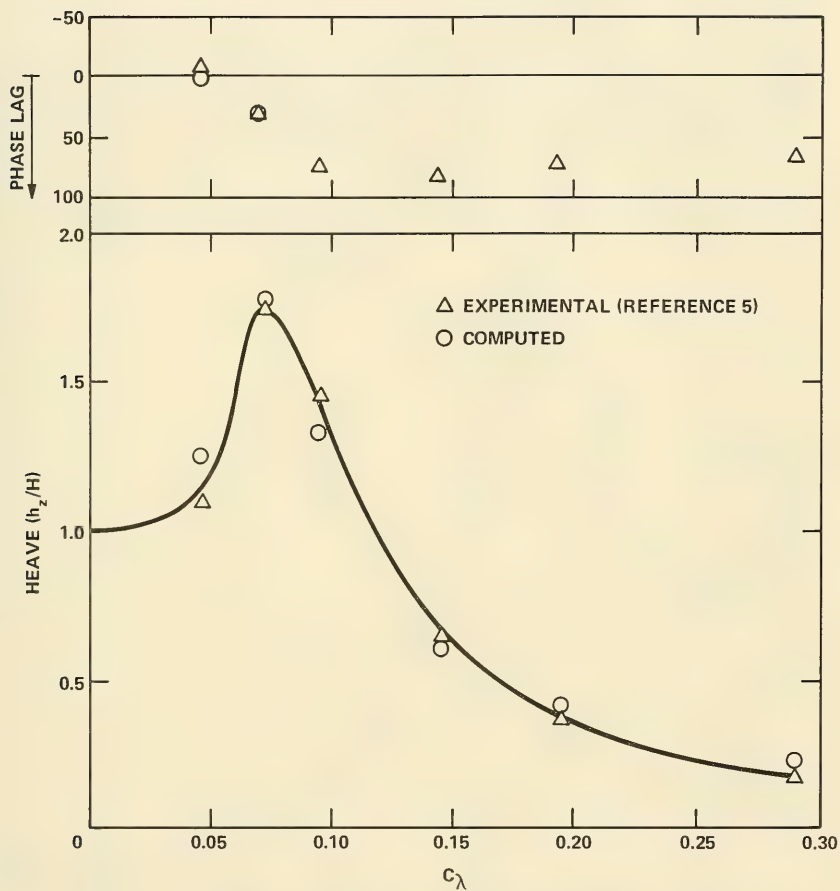


Figure 11 – Heave Response for 20-Degree Deadrise Model at $V/\sqrt{L} = 6.0$

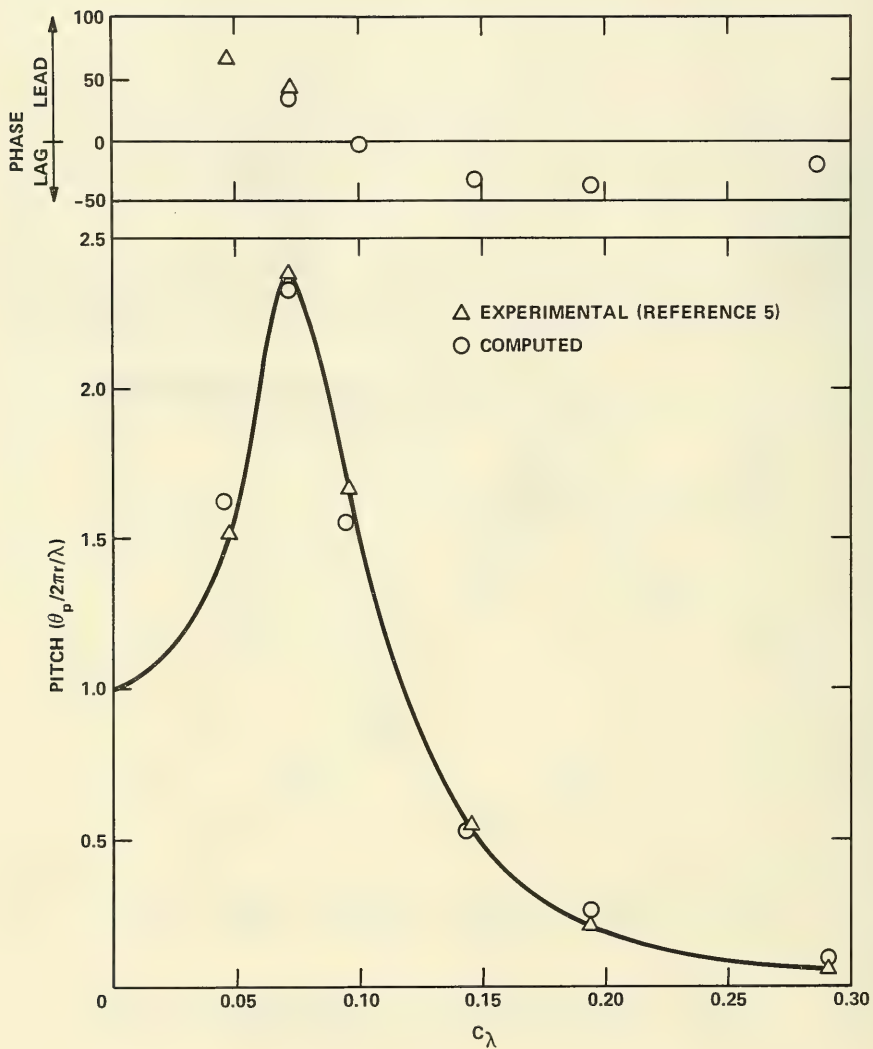


Figure 12 – Pitch Response for 20-Degree Deadrise Model at $V/\sqrt{L} = 6.0$

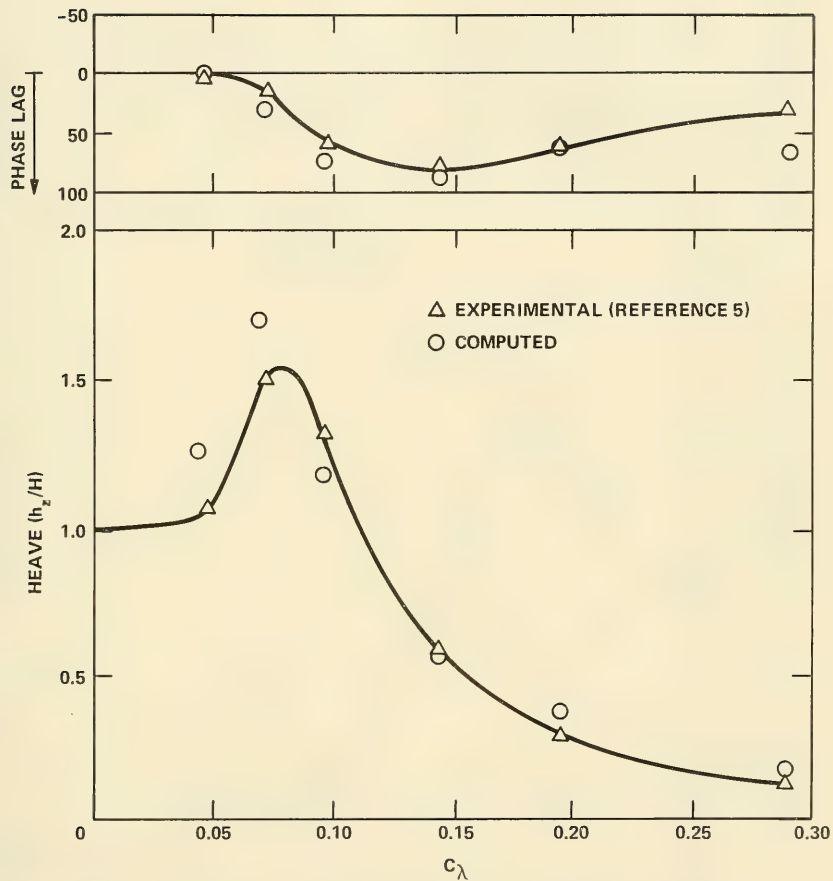


Figure 13 – Heave Response for 30-Degree Deadrise Model at $V/\sqrt{L} = 6.0$

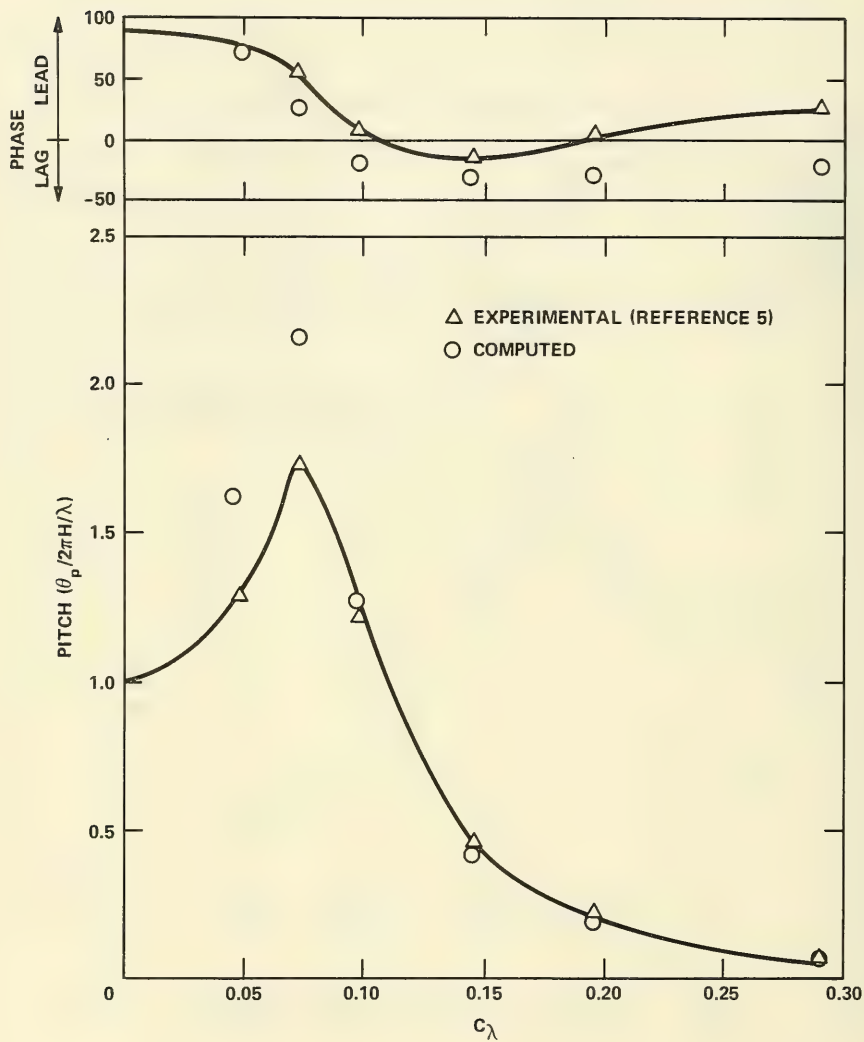


Figure 14 – Pitch Response for 30-Degree Deadrise Model at $V/\sqrt{L} = 6.0$

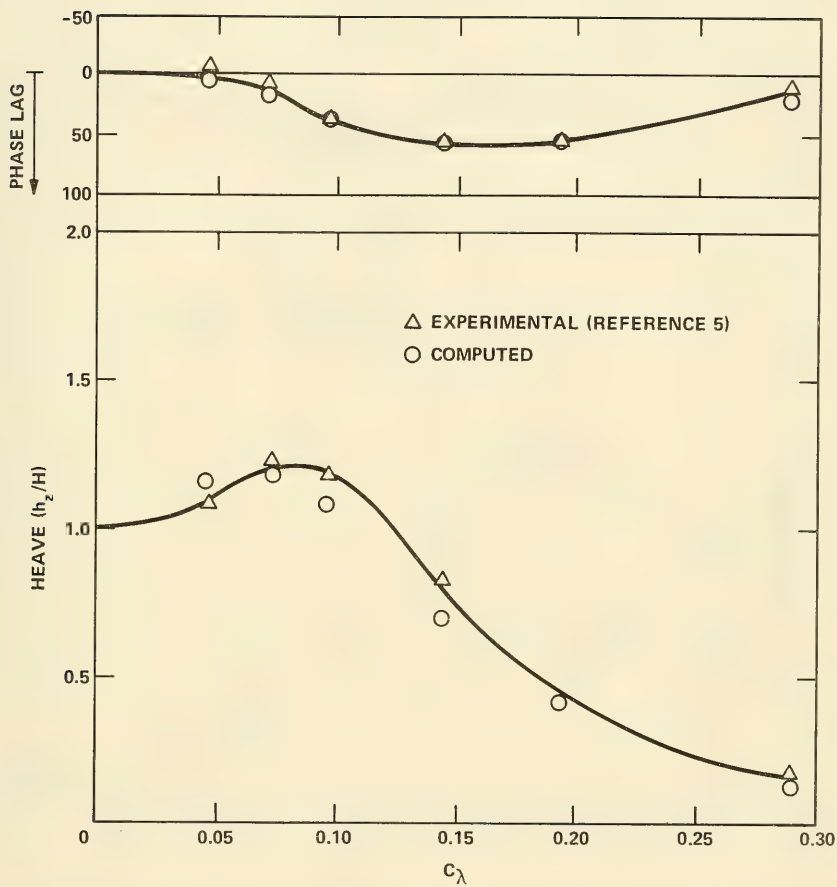


Figure 15 – Heave Response for 20-Degree Deadrise Model at $V/\sqrt{L} = 4.0$

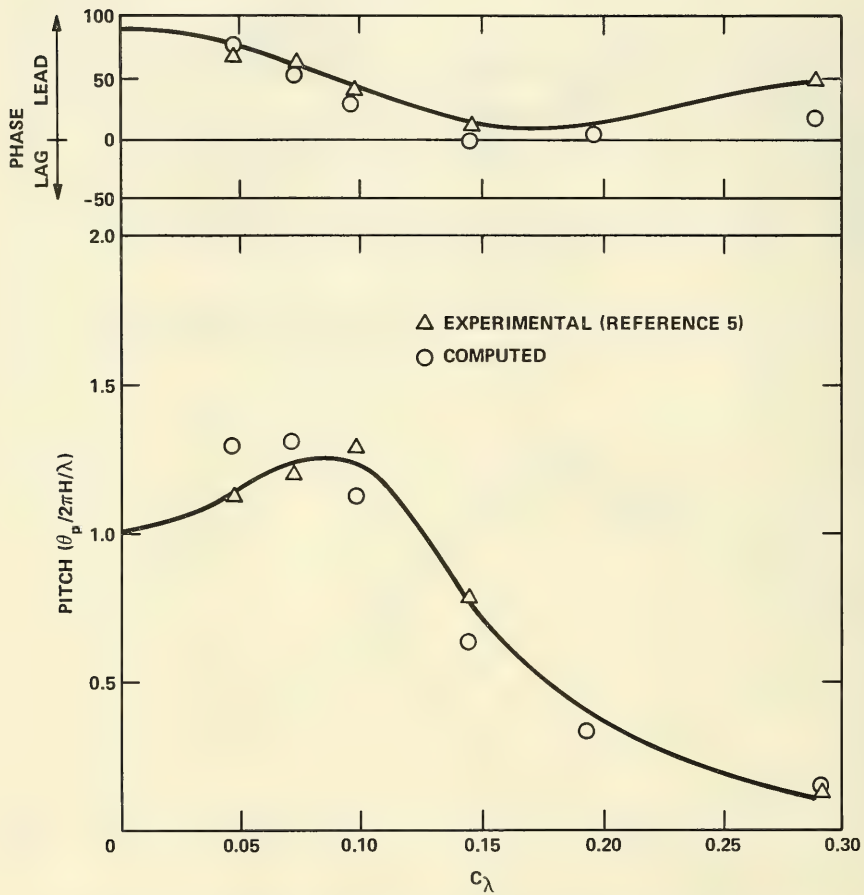


Figure 16 – Pitch Response for 20-Degree Deadrise Model at $V/\sqrt{L} = 4.0$

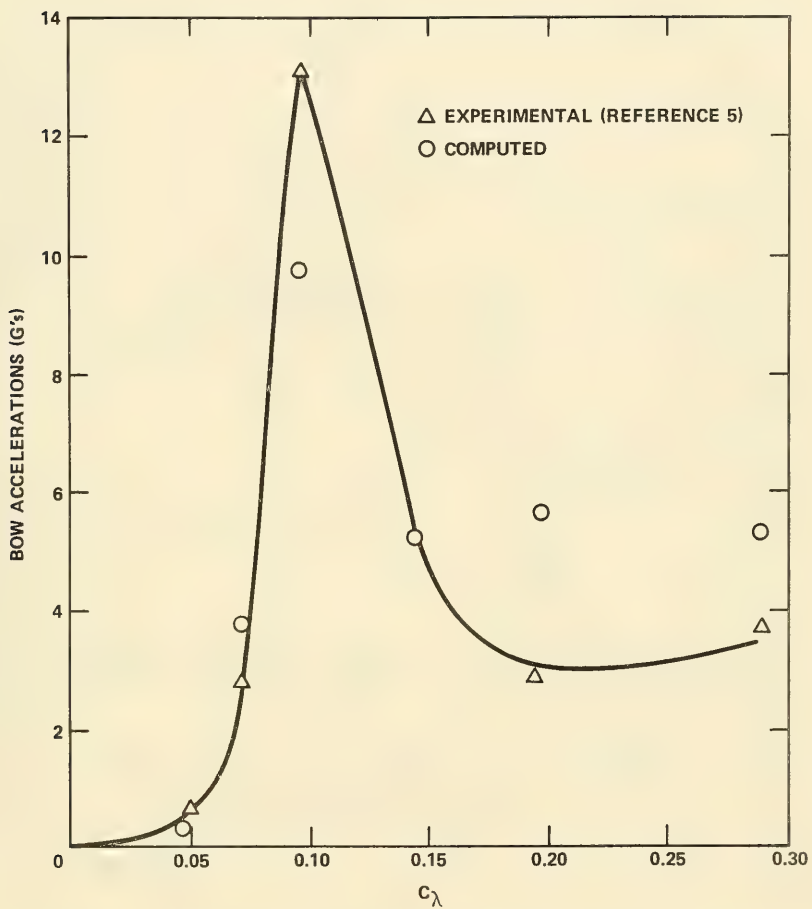


Figure 17 - Bow Acceleration for 10-Degree Deadrise Model at $V/\sqrt{L} = 6.0$

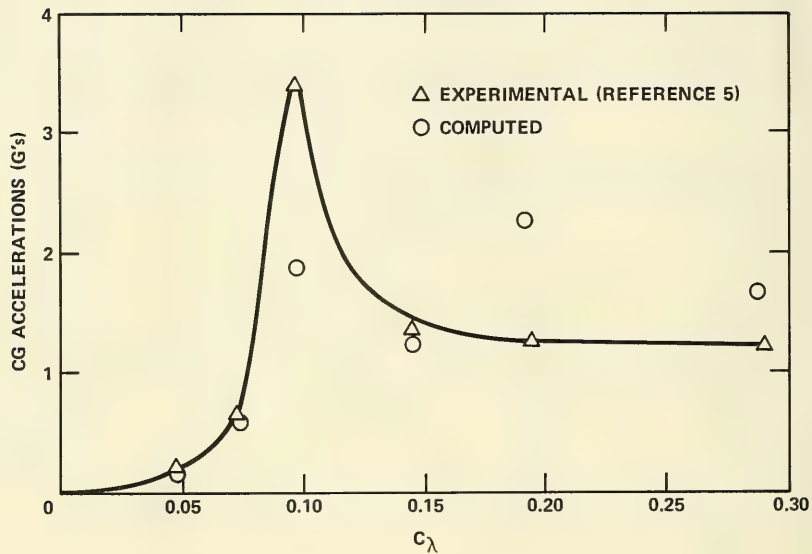


Figure 18 – Center of Gravity Acceleration for 10-Degree Deadrise Model at $V/\sqrt{L} = 6.0$

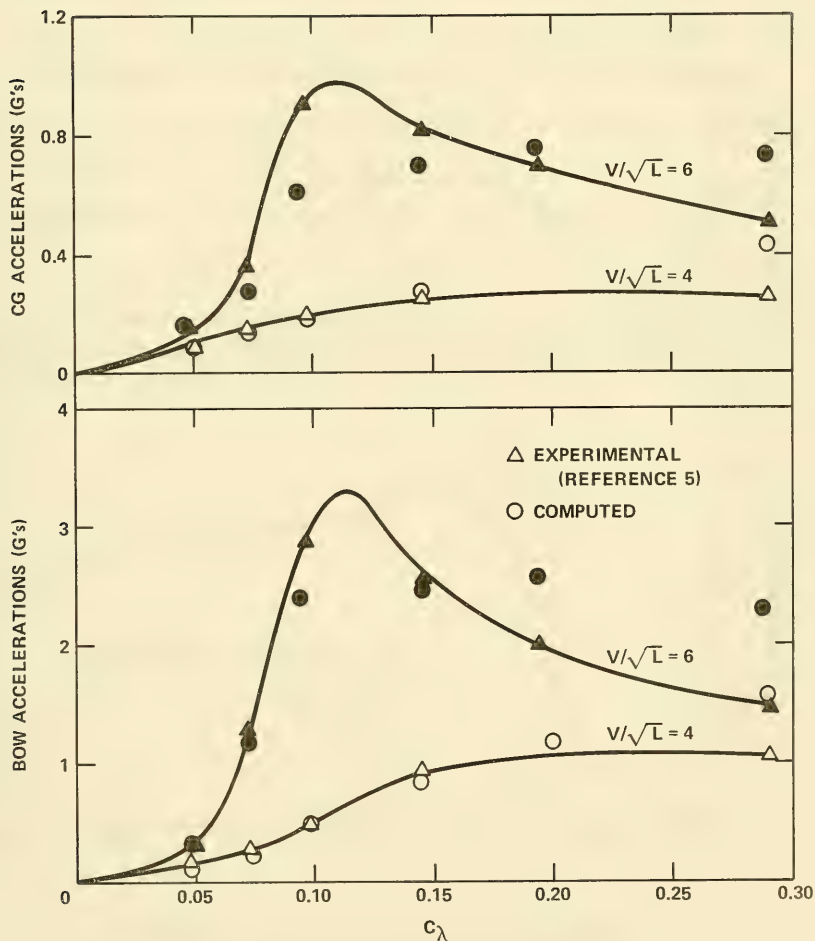


Figure 19 – Bow and Center of Gravity Accelerations for 20-Degree Deadrise Model at $V/\sqrt{L} = 4.0$ and $V/\sqrt{L} = 6.0$

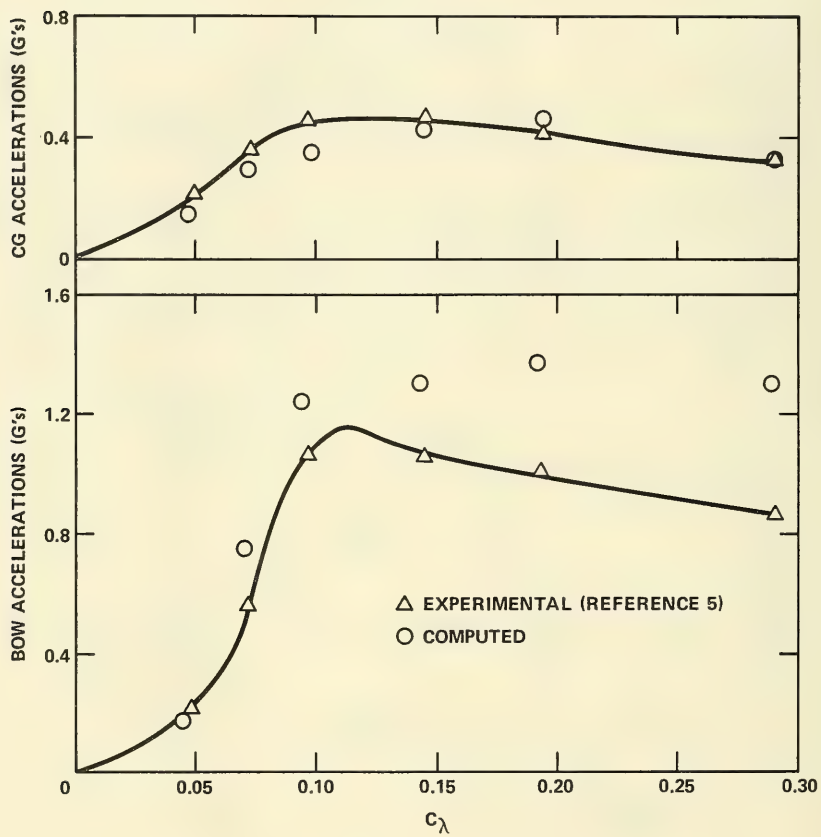


Figure 20 – Bow and Center of Gravity Accelerations
for 30-Degree Deadrise Model at $V/\sqrt{L} = 6.0$

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3. Martin, M., "*Theoretical Predictions of Motions of High-Speed Planing Boats in Waves*," DTNSRDC Report 76-0069 (Apr 1976).
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APPENDIX A

EVALUATION OF HYDRODYNAMIC FORCE AND MOMENT INTEGRALS

The hydrodynamic force the craft experiences in the vertical direction as derived in the text is:

$$F_z = - \int_{\ell} \left\{ m_a \dot{V} - U \frac{\partial m_a V}{\partial \xi} + \dot{m}_a V + C_D \rho b V^2 \right\} \cos \theta \, d\xi + \int_{\ell} a \rho g A \, d\xi$$

where $U = \dot{x}_{CG} \cos \theta - (\dot{z} - w_z) \sin \theta$

and

$$V = \dot{x}_{CG} \sin \theta + (\dot{z} - w_z) \cos \theta - \dot{\theta} \xi$$

Another force acting in the vertical direction is the weight of the craft.

The first two terms of the integral are evaluated by making the substitutions

$$\begin{aligned} \dot{V} &= \ddot{x}_{CG} \sin \theta - \ddot{\theta} \xi + \ddot{z}_{CG} \cos \theta - \dot{w}_z \cos \theta \\ &\quad + \dot{\theta} (\dot{x}_{CG} \cos \theta - \dot{z}_{CG} \sin \theta) + w_z \dot{\theta} \sin \theta \end{aligned}$$

$$\frac{\partial V}{\partial \xi} = -\dot{\theta} - \frac{\partial w_z}{\partial \xi} \cos \theta$$

$$\frac{\partial U}{\partial \xi} = \frac{\partial w_z}{\partial \xi} \sin \theta$$

$$\frac{dw_z}{dt} = \dot{w}_z - U \frac{\partial w_z}{\partial \xi}$$

and noting that

$$\int_{\ell} UV \frac{\partial m_a}{\partial \xi} \, d\xi = -UV m_a \Big|_{\text{stern}} - \int_{\ell} m_a \frac{\partial UV}{\partial \xi} \, d\xi$$

Using the previously described substitutions, the force becomes

$$\begin{aligned}
F_z = & \left\{ -M_a \cos \theta \ddot{z}_{CG} - M_a \sin \theta \ddot{x}_{CG} + Q_a \ddot{\theta} + M_a \dot{\theta} (\dot{z}_{CG} \sin \theta - \dot{x}_{CG} \cos \theta) \right. \\
& + \int_{\ell} m_a \frac{d w_z}{dt} \cos \theta d\xi - \int_{\ell} m_a w_z \dot{\theta} \sin \theta d\xi \\
& - \int_{\ell} m_a V \frac{\partial w_z}{\partial \xi} \sin \theta d\xi + \int_{\ell} m_a U \frac{\partial w_z}{\partial \xi} \cos \theta d\xi \\
& \left. - UV m_a \Big|_{\text{stern}} - \int_{\ell} V \dot{m}_a d\xi - \rho \int_{\ell} C_{D,c} b V^2 d\xi \right\} \cos \theta \\
& + \int_{\ell} a \rho g A d\xi
\end{aligned}$$

where $M_a = \int_{\ell} m_a d\xi$

and

$$Q_a = \int_{\ell} m_a \xi d\xi$$

This is essentially the form in which the integrals have been computed in the program.

The rate of change of the sectional added mass in the third term of the integral expression is derived by relating it to the rate of change of depth of fluid penetration of the section. The added mass of a section is assumed to be equal to

$$m_a = k_a \pi/2 \rho b^2$$

for which the time derivative is

$$\dot{m}_a = k_a \pi \rho b \dot{b}$$

where b is the instantaneous half-beam of the section, and k_a is an added-mass coefficient, assumed to be constant. A value of $k_a = 1.0$ was used in the computations contained in this report. For sections with constant deadrise, which is an imposed limitation of this work, the half-beam is related to the depth of penetration by

$$b = d \cot \beta$$

where d is depth of penetration, and β is deadrise angle.

Taking into account the effect of water pileup, the effective depth of penetration d_e is, according to Wagner

$$d_e = \pi/2 d$$

and

$$b = d_e \cot \beta = \pi/2 d \cot \beta$$

where $\pi/2$ is the factor by which the wedge immersion is increased by the pileup. Using this expression for the half-beam, the rate of change of sectional added mass becomes

$$\dot{m}_a = k a \pi \rho b (\pi/2 \cot \beta) \dot{d}$$

This expression is valid for penetration of the section up to the chine. When the immersion exceeds the chine, the sectional added mass is assumed to be constant, i.e.,

$$\begin{aligned} m_a &= k \pi/2 \rho b_{\max}^2 \\ \dot{m}_a &= 0 \end{aligned}$$

where $b_{\max}$ is the half-beam at chine.

The submergence of a section in terms of the motions is given by

$$h = z - r$$

where $z = z_{CG} - \xi \sin \theta + \zeta \cos \theta$

$$r = r_o \cos \{k(x_{CG} + \xi \cos \theta + \zeta \sin \theta) + \omega t\}$$

For wavelengths which are long in comparison to the draft and for small wave slopes, the immersion of a section measured perpendicular to the baseline is approximately

$$d \approx \frac{z - r}{\cos \theta - \nu \sin \theta}$$

where ν = wave slope

The rate change of submergence d is given by

$$\dot{d} = \frac{\dot{z} - \dot{r}}{\cos \theta - v \sin \theta} + \frac{(z - r)}{(\cos \theta - v \sin \theta)^2} \cdot \frac{\partial (\cos \theta - v \sin \theta)}{\partial t}$$

Since immersion $(z - r)$ is always small in the valid range of the previously described expression, the relationship can be further simplified to

$$\dot{d} \approx \frac{\dot{z} - \dot{r}}{\cos \theta - v \sin \theta}$$

and

$$\dot{m}_a \approx k_a \pi \rho b (\pi/2 \cot \beta) \frac{(\dot{z} - \dot{r})}{\cos \theta - v \sin \theta}$$

The expansion of the integral expression for the hydrodynamic moment in pitch follows the procedure used for the vertical force. The results are summarized as follows

$$\begin{aligned} F_\theta = & -I_a \ddot{\theta} + Q_a \cos \theta \ddot{z}_{CG} - Q_a \dot{\theta} (\dot{z}_{CG} \sin \theta - \dot{x}_{CG} \cos \theta) \\ & - \int_{\ell} m_a \cos \theta \frac{dw_z}{dt} \xi d\xi + \int_{\ell} m_a \dot{\theta} \sin \theta w_z \xi d\xi \\ & + \int_{\ell} V \dot{m}_a \xi d\xi + \int_{\ell} \rho C_D b V^2 \xi d\xi \\ & + m_a UV \xi \Big|_{\text{stern}} + \int_{\ell} m_a V U d\xi \\ & + \int_{\ell} m_a V \frac{\partial w_z}{\partial \xi} \sin \theta \xi d\xi \\ & - \int_{\ell} m_a U \frac{\partial w_z}{\partial \xi} \cos \theta \xi d\xi \\ & + \int_{\ell} a \rho g A \cos \theta \xi d\xi \end{aligned}$$

The only additional moments are the buoyancy moments. All other moments are considered to be zero for the specific problem considered in this report.

APPENDIX B

COMPUTER PROGRAM DESCRIPTIONS

OVERVIEW

The equations of motions developed in the previous sections of this report have been solved by means of digital computer programs. Two major programs have been developed: the first (MAIN) solves the equations of motion using the Runge-Kutta-Merson integration algorithm and generates time histories that are stored on the system disk. The second (PLTHSP) generates California Computer Products Company (CALCOMP) pen plots from the disk files. All programs were designed to operate on the Control Data Corporation computer system, located at the David W. Taylor Naval Ship Research and Development Center in Carderock, Md.

Descriptions of input data required to execute the programs, job control cards, and programs follow. Sufficient detail is presented for this appendix to serve as a manual for use and maintenance.

JOB CONTROL CARDS FOR PROGRAM MAIN

Job control cards for program MAIN which computes time histories of the motion variables, are described as follows. If CALCOMP plots are not desired, TAPES need not be cataloged.

Job Control Language Card:

Comment

Job Card	Standard facility card
Charge Card	Standard facility card
REQUEST,TAPE9,*PF.	Reserves space for CALCOMP plot data
REQUEST,TAPE2,*PF.	Print output file 1 request
REQUEST,TAPE4,*PF.	Print output file 2 request
ATTACH,BINAR,SEFZARNICKNEWB, ID=XXXX.	Attaches binary run file
ATTACH,NSRDC.	Attaches library routines
LDSET(LIB=NSRDC).	Loads library routines
BINAR.	Loads and executes run file
REWIND,TAPE2. REWIND,TAPE4.	Rewinds time-history files for printing
COPY(TAPE2,OUTPUT)	Prints time-history file
COPY(TAPE4,OUTPUT)	Prints time-history file

Job Control Language Card:

Comment

CATALOG,TAPE9, SEFZARNICKDATA., ID=XXXX. Catalogues file for plot.
(SEFZARNICKDATA CAN BE ANY NAME)

7/8/9 END OF RECORD

DATA CARDS (1-5)

6/7/8/9 END OF FILE

INPUT DATA CARDS FOR PROGRAM MAIN

Input data used by program MAIN are read from data cards in NAMELIST and in standard format. A description of the FORTRAN symbols appearing in NAMELIST follows. For simplicity in the text that follows, it is assumed that NAMELIST input occupies only one card. More cards can be used if necessary.

Card 1(NAMELIST FORMAT, / /)

A	The absolute error for KUTMER (six values)
NPRINT	If=1, print normal output If=2, matrix, inverse matrix, F-column matrix, and KUTMER results If=3, integral results If=4, calculated values constant for given input values
NPLOT	If=0, no plot If=1, printer plot of results
END	Number of runs to be made
W	Weight of craft in pounds
BL	Boat length in feet
TZ	Thrust component in z direction
TX	Thrust component in x direction
XECG	Distance from center of gravity to center of pressure for drag force in feet
XP	Moment arm of propeller thrust
XD	Distance from center of gravity to center
DRAG	Friction for drag force
RO	Wave height
LAMBDA	Wavelength
RG	Radius of gyration in feet
T	Propeller thrust in pounds
GAMMA	Propeller thrust angle in degrees

Card 1 (continued)

ECG	Longitudinal center of gravity
NCG	Vertical center of gravity, nondimensionalized by ship length
KAR	Added-mass coefficient
BETA(I)	Dead-rise angle in degrees
EST(I)	Station position in feet
NUM	Number of stations
XA	Initial time
XE	Stop time
HMIN	Minimum step size
HMAX	Maximum step size
EPS	Error criterion

Card 2 (Format 8F10.0)

(X(I),I=1,6)	Initial conditions
X(1)	Velocity
X(2)	Z
X(3)	θ
X(4)	X
X(5)	Z
X(6)	θ degrees

Card 3 (8F10.0)

START	Time to turn on (RMP) function (see page 48)
RISE	Duration of RMP

Card 4 (8F10.0)

TME	Time at which integration interval is to be changed*
HMX	New maximum interval size after TME
HMN	New minimum interval size for KUTMER to subdivide

*If this option is not used set TME to stop time on run.

Card 5 (8F10.0)

PERCNT Percentage of boat length subtracted from longitudinal center of gravity to obtain X - point where acceleration computations are made

JOB CONTROL CARDS FOR PROGRAM PLTHSP

Job control cards for program PLTHSP which generates CALCOMP plots of time histories computed by program MAIN are described in this section.

Job Control Language Card:

Comment

Job Card	Standard facility card
Charge Card	Standard facility card
REQUEST,TAPE7,HI.	Tape for CALCOMP plot data
VSN(TAPE7=CK0323).	Volume serial number of tape for CALCOMP plot
ATTACH,CALC936.	Attaches CALCOMP library routine
ATTACH,BINAR,SEFZARNICKPLOTB, ID=XXXX.	Attaches plot program run file
LDSET(LIB=CALC936)	Loads CALCOMP library routines
BINAR.	Runs plot program
7/8/9 END OF RECORD	
DATA CARDS	
6/7/8/9 END OF FILE	

INPUT DATA CARDS FOR PROGRAM PLTHSP

Two or three data cards are made ready by PLTHSP, depending on the options selected. Standard input format is employed. A description of the necessary data cards follows.

Card 1 (8F10.0 Format)

XAXIS	Length of x axis in inches
YAXISP	Height of pitch component axis in inches
YAXISH	Height of heave component axis in inches
HT	Height of lettering in inches

Card 2 (I10 Format)

IA	If = 0, no plots for bow acceleration and center of gravity acceleration
	If = 1, plots previously mentioned information

Card 3 (8F10.0 Format) - Only Necessary If IA = 1.

YAXISB	Height of bow acceleration axis in inches
YAXISC	Height of CG acceleration axis in inches

PROGRAM MAIN

Program MAIN reads all necessary input data from cards, sets up initial values, computes constants, calls KUTMER to determine the state variables at TIME for the period from XA to XE in increments of HMAX. A table state variables is created for every PTIME-th value. The values for λ/H and $\theta_p/2\pi H/\lambda$ are calculated and printed. If the plot option is on, a printer plot will be produced.

Subroutine COMPUT(X)

This routine computes pitch moment NL and lift force FL, excluding added mass terms, using values of integrals computed in subroutine FUNCT. The argument X contains the state vector.

Subroutine DAUX

This subroutine is called from KUTMER or EULER. It determines the values of m_a , b , and $b1^*$, based on the following equations

$$h_w(I) = z_{CG} - \xi(I) \sin \theta + \zeta(I) \cos \theta - r(I)$$

$$\text{where } r(I) = r_o \cos k [x_{CG} + \xi(I) \cos \theta + \zeta(I) \sin \theta + ct]$$

Then for

$$h_w(I) > 0 ,$$
$$d(I) = \frac{h_w(I)}{\cos \theta - (I) \sin \theta}$$

$$\text{where } V(I) = -r_o k \sin \theta [x_{CG} + \xi(I) \cos \theta + (I) \sin \theta + ct]$$

If

$$d(I) \geq b_m(I) \tan (\beta(I) \geq 2/\pi)$$

set

$$m_a(I) = m_{amax}(I)$$

$$b(I) = b_m(I)$$

$$b1(I) = 0$$

$$m_{amax}(I) = k(I) (\rho/2) \pi b_m^2(I)$$

If

$$d(I) < b_m(I) \tan \{\beta(I)\} (2/\pi)$$

set

$$b(I) = d(I) \cot \{\beta(I)\} (\pi/2)$$

$$b1(I) = b(I)$$

$$m_a(I) = k_a(I) (\rho/2) \pi b^2(I)$$

for

$$h_w(I) \leq 0 ;$$

$$m_a(I) = 0, \quad b(I) = 0, \quad b1(I) = 0$$

This subroutine then calls FUNCT which in turn calls COMPUT to determine the values of N_L and F_L , the lift force and moment. The values of N_L and F_L are used to compute the following

$$F_1 = T_x + F_L \sin \theta - D \cos \theta$$

$$F_2 = T_z + F_L \cos \theta + D \sin \theta + W$$

$$F_3 = N_L - D_{xd} + T_{xp}$$

*b1 array is set up for integrations for portion of hull for which chine is not immersed.

The mass inertia matrix is

$$A_{11} = M + M_a \sin^2 \theta$$

$$A_{12} = M_a \sin \theta \cos \theta$$

$$A_{13} = -Q_a \sin \theta$$

$$A_{21} = A_{12}$$

$$A_{22} = M + M_a \cos^2 \theta$$

$$A_{23} = -Q_a \cos \theta$$

$$A_{31} = A_{13}$$

$$A_{32} = A_{23}$$

$$A_{33} = I + I_a$$

The matrix is inverted by the system routine MATINS. The inverted matrix is then used to solve the following equations which determine the state vectors.

$$\ddot{x}_{CG} = A_{11}^{-1} F_1 + A_{12}^{-1} F_2 + A_{13}^{-1} F_3$$

$$\ddot{z}_{CG} = A_{21}^{-1} F_1 + A_{22}^{-1} F_2 + A_{23}^{-1} F_3$$

$$\ddot{\theta} = A_{31}^{-1} F_1 + A_{32}^{-1} F_2 + A_{33}^{-1} F_3$$

Subroutine FUNCT (X)

This routine evaluates various integrals appearing in the force and moment mathematical models. The integrals are evaluated, using a trapezoidal integration algorithm. The argument x contains the state vector. A list of integrals that are evaluated is presented.

$$\int_{\ell} m_a d\xi$$

$$\int_{\ell} m_a \xi d\xi$$

$$\int_{\ell} m_a \xi^2 d\xi$$

$$\int_{\ell} m_a U V d\xi$$

$$\int_{\ell} m_a w_z d\xi$$

$$\int_{\ell} m_a w_z \xi d\xi$$

$$\int_{\ell} m_a \frac{dw_z}{dt} d\xi$$

$$\int_{\ell} m_a \frac{dw_z}{dt} \xi d\xi$$

$$\int_{\ell} m_a V \frac{\partial w_z}{\partial \xi} d\xi$$

$$\int_{\ell} m_a V \frac{\partial w_z}{\partial \xi} \xi d\xi$$

$$\int_{\ell} m_a U \frac{\partial w_z}{\partial \xi} d\xi$$

$$\int_{\ell} m_a U \frac{\partial w_z}{\partial \xi} \xi d\xi$$

$$\int_{\ell} m_a V d\xi$$

$$\int_{\ell} m_a V \xi d\xi$$

$$\int_{\ell} b V^2 d\xi$$

$$\int_{\ell} b V^2 \xi d\xi$$

$$\int_{\ell} b \left(h - \frac{b}{2} \tan \beta \right) d\xi$$

$$\int_{\ell} b \left(h - \frac{b}{2} \tan \beta \right) \xi d\xi$$

Subroutine INPUT

This routine reads in NAMELIST/HSP/ which contains the initial data concerning the craft and sea conditions pertinent to all the runs to be made. It is set up so that most of the data are given default values by means of data statements in subroutine INPUT. These data statements can be overridden during execution by reading values in on cards. For further explanation of the specific variables see section on the input data cards.

This routine also "initializes" constant such as π , ρ , and g . It uses the input values to calculate the keel profile and planform arrays, NO and BM, wave constants, system mass and inertia, and maximum mass and depth of chine at each station.

Subroutine KUTMER (NEQS, TIME, HMAX, X, EPSE, A, HMIN, FIRST)

This is a Runge-Kutta-Merson integration routine that is capable of changing the size of the interval over which it integrates to meet specified error criteria. It is therefore an

accurate method for a system that may oscillate more rapidly than the initial integration interval. A minimum step size prevents the routine from subdividing the interval indefinitely.

The input arguments are:

NEQS	Number of dependent variables in the x array
TIME	Actual time (independent variable)
HMAX	Increment for which the solution is to be returned
X	Vector of dependent variables
EPSE	Relative error criteria specified for each component of x and used for the components of x less than the absolute value of A
A	Absolute error criteria
HMIN	Minimum step size allowed
FIRST	Set to zero on first call; a value of 1 is assigned by KUTMER on subsequent calls for which the error criteria are satisfied, otherwise a value of 2 is assigned

Subroutine PLOT2 (F, FMIN, FMAX, NVAR, NFUN, N1, N, XO, DELX)

Data stored in the two-dimensional array F are plotted, using the printer by subroutine PLOT2. As many as 26 different functions, having evenly spaced abscissa values, can be plotted. The output is written on Unit 6. A description of variables follows.

F	Array containing data to be plotted; the Jth point of the Ith function is stored in F(I,J)
FMIN	An array of minimum functional values; the minimum of the Ith function is stored in FMIN(I)
FMAX	Same as FMIN only for maximum values
NVAR	An array of titles for each function to be plotted
NFUN	Number of functions to be plotted
N1	First dimension of array F
N	Number of points to be plotted
XO	First abscissa value
DELX	Abscissa increment

Subroutine PLOTER (FX, XA, HMAX, LAMBDA, IB, NWAIVE)

The routine initializes various values required to generate printer plots and computes pitch-and-heave ratios. The printer plots that are generated consists of pitch-and-heave time histories. A description of input variables follows.

FX	A two-dimensional array, containing time histories to be plotted
XA	Initial time
HMAX	Time-interval increment; time interval between values in FX is given by HMAX*PTIME
LAMBDA	Wavelength
IB	Number of values to be plotted
NWAVE	Position in FX at which wave is completely turned on

Function RMP (T, START, RISE)

The RMP is a function that calculates a value between 0 and 1 corresponding to time T, based on a straight line from time START with a value of 0 to time START plus RISE with a value of 1. It is used to lower the initial wave amplitude to avoid large transients at start of the computations.

The arguments are:

T	Actual time
START	Time at which to begin the ramp from 0 to 1
RISE	Duration of rise from 0 to 1

The function reaches the value 1 at time START plus RISE, if the rise is 0.0, RMP will return a value of 0.5.

Subroutine TRAP (F, DX, NPTS, ANS)

This routine performs the evaluation of an integral using a trapezoidal approximation.

The argument variables are defined as follows:

F	Array of integrand values
DX	Increments at which F is evaluated
NPTS	Number of values in F
ANS	Result, which is equal to

$$DX \left\{ \sum_{i=1}^{NPTS} F(i) - 0.5 [F(1) + F(NPTS)] \right\}$$

PROGRAM PLTHSP

This program uses a data file created by program MAIN to create CALCOMP plots. The data are read from logical Unit 9 and are rewritten on Unit 7 for CALCOMP input. Program PLTHSP sets the tape output unit equal to 7 and calls SUBROUTINE CALPHI to execute the plot procedures.

Subroutine CALPLT

This subroutine manages all the I/O operations and performs the necessary calculations required to generate the plots. After reading the card data (two or three cards) subroutine READT is called to read the data file (Tape 9) created by program MAIN. The CALCOMP initializing routines are called next, after which a call to subroutine ESCALE calculates the necessary scaling factors. Subroutine EXAXIS is called next to determine the placement of the plot tick marks and identifying digits. The CALCOMP plot-generation subroutines are now called and, depending on the option defined by the IA parameter on card 2, plots of pitch and heave at the bow and CG location are generated as functions of time if IA = 1.

Subroutine EAXIS

The subroutine is analogous to the CALCOMP AXIS routine. The only exception is that the tick marks are not necessarily inch, and the height of the characters is defined by the input parameter HT. Function NDIGIT is called to determine the number of digits necessary to print an even increment of the plots functions on the axis.

Subroutine ESCALE, ADJUST, and FUNCTION UNIT

These subroutines find the scale to be used on the plot axis. Function UNIT is called to determine the axis increment size after which subroutine ADJUST is called to extend the minimum (AMIN) and maximum (AMAX) values so that they are even multiples of the axis increments.

FUNCTION NDIGIT

This function finds the number of digits necessary to print even increments of the function on the axis. Both the number of places in the entire number (NDIGIT) and the number of decimal places (ND) are determined, after which the value of each increment on the axis (ANUM) is calculated.

Subroutine READT

This subroutine reads the data file created by program MAIN. Data file records are read until the message end of file is encountered. Each record is read in the same format as it was written in MAIN. The information is printed to allow the user to inspect the created file.

LISTING OF COMPUTER PROGRAM FOR MOTION COMPUTATIONS

	PROGRAM MAIN(INPUT,OUTPUT,TAPE5=INPUT,TAPE6=OUTPUT,TAPE3=512,	MAIN	2
	TAPE2=512,TAPE4=512,TAPE9)	MAIN	3
C	REAL IT,K,LAMBDA,M,MA,MMAX,N,NCG,NU,MASS,NL,IA,KAR	MAIN	4
	INTEGER END	MAIN	5
C		MAIN	6
	DIMENSION X(6),FX(2,400)	MAIN	7
C		MAIN	8
	COMMON /CONST/ NCG,ECG,PI,DPR,RPD,GRAVTY,RHO,K,NUM,MA(120),CD,TA,	MAIN	9
	B(120),BETA,HW(120),TZ,URAG,W,XD,T,XP,M,IT,	MAIN	10
	DELTA S,TX,EST(120),C,RO,KAR,MMAX(1 0),TEST(120),	MAIN	11
	N(120),PHALF	MAIN	12
	COMMON /SHIP/ MASS,CINT,QA,CE,CE2,CE3,DMU,EDMU,E2DMU,E3DMU,BF,BMM,	MAIN	13
	NL,FL,IA,E(120)	MAIN	14
	COMMON /IN/ BM(120),B1(120),VELIN	MAIN	15
	COMMON/OUT/NPRINT,NPLOT,END	MAIN	16
	COMMON/TERMS/T1,T2,T3,T4,T5,T6,T7,T8	MAIN	17
	COMMON /SEAWAVE/ START,RISE,RAMP	MAIN	18
	COMMON /INTER/ II,KTT(10),DIFF(10)	MAIN	19
	COMMON /IN2/ NO(120),XA,XE,HMAX,HMIN,A(6),EPSE(6),LAMBDA	MAIN	20
	COMMON /ACCEL / XACCL,BWACL,CGACL,BL	MAIN	21
C		MAIN	22
	CALL INPUT	MAIN	23
C		MAIN	24
	COMPUTE INTEGRATION INTERVAL INFORMATION	MAIN	25
C		MAIN	26
	NLESS = NUM-1	MAIN	27
	I = 1	MAIN	28
	II = 1	MAIN	29
	DIFFER = EST(I+1)-EST(I)	MAIN	30
	KTT(II) = 1	MAIN	31
	DIFF(II) = DIFFER	MAIN	32
	DO 25 I=2,NLESS	MAIN	33
	DIFFER= EST(I+1)-EST(I)	MAIN	34
	KTT(II) = KTT(II)+1	MAIN	35
	IF(DIFFER.NE.DIFF(II))GO TO 24	MAIN	36
	GO TO 25	MAIN	37
24	II = II+1	MAIN	38
	KTT(II) = 1	MAIN	39
	DIFF(II) = DIFFER	MAIN	40
25	CONTINUE	MAIN	41
	KTT(II) = KTT(II)+1	MAIN	42
C	*** CHECK IF NUMBER OF INTERVALS EXCEEDS DIMENSION	MAIN	43
	IF (II.GT.10) WRITE(6,28) (KTT(I),DIFF(I),I=1,II)	MAIN	44
	IF(II.GT.10) STOP 4	MAIN	45
C	*** POINT AT WHICH MULTIPLE RUNS START	MAIN	46
	8 CONTINUE	MAIN	47
	TIME=XA	MAIN	48
	KOUNT=1	MAIN	49
	END=END-1	MAIN	50
	WRITE(6,39)	MAIN	51
	39 FORMAT(1H1)	MAIN	52
C	***** READ IN INITIAL CONDITIONS	MAIN	53
C	X(1) = VELOCITY, X(2) = Z DOT, X(3) = THETA DOT	MAIN	54
C	X(4) = X, X(5) = Z, X(6) = THETA	MAIN	55
C	THETA IS PEAD IN DEGREES THEN CONVERTED TO RADIANS IN PROGRAM	MAIN	56
C		MAIN	57
	READ(5,10) (X(I),I=1,6)	MAIN	58
C		MAIN	59
		MAIN	60

C	DATA , USED IN RAMP FUNCTION, TO TURN ON WAVE	MAIN	61
	READ(5,10)START,RISE	MAIN	62
C		MAIN	63
	10 FORMAT(8F10.4)	MAIN	64
C	* * * * * WRITE OUT THE INPUT VALUES	MAIN	65
	WRITE(6,19) START,RISE,KAR	MAIN	66
	19 FORMAT(" START = ",F10.4,/, " RISE = ",F10.4,/, " KAR = ",F10.4,/,	MAIN	67
	0.4)	MAIN	68
C		MAIN	69
C	TME IS THE TIME AT WHICH THE INTEGRATION INTERVAL IS	MAIN	70
C	TO BE CHANGED	MAIN	71
C	HMX IS THE NEW MAXIMUM INTERVAL SIZE AFTER TIME TME	MAIN	72
C	HMN IS THE NEW MINIMUM INTERVAL SIZE FOR KUTMER TO SUB-DIVIDE	MAIN	73
C	THE MAXIMUM INTERVAL UP TO	MAIN	74
C	IF THIS OPTION IS NOT USED SET TME TO THE STOP TIME OF THE RUN	MAIN	75
C		MAIN	76
	READ(5,10) TME,HMX,HMN	MAIN	77
	WRITE(6,11) TME,HMAX,HMX,HMIN,HMN	MAIN	78
	11 FORMAT(* AT TIME *,F7.2,* THE MAXIMUM INTERVAL SIZE FOR INTEGRATION	MAIN	79
	*ON WILL BE CHANGED FROM *,F10.4,* TO *,F10.4,/,	MAIN	80
	** AND THE MINIMUM SIZE FOR HALVING CHANGES FROM *,F10.4,	MAIN	81
	* * TO *,F10.4)	MAIN	82
C	ADJUST THE TIME FOR CHANGE OF INTEGRATION INTERVAL	MAIN	83
C	FOR CHECK AGAINST TIME IN THE INTEGRATION LOOP	MAIN	84
C	TM = TME-(HMAX/2.)	MAIN	85
C	SET SWITCH FOR CALCULATION OF PITCH AND HEAVE RATIOS	MAIN	86
C	ON NEXT CALL TO PLOTTER	MAIN	87
	IPT = 0	MAIN	88
	IF(TME.EQ.XE) IPT = 1	MAIN	89
C		MAIN	90
	READ(5,10) PERCNT	MAIN	91
	XACCL = ECG-PERCNT*BL	MAIN	92
	WRITE(6,12) PERCNT,XACCL	MAIN	93
	12 FORMAT(* THE X USED FOR THE BOW AND CG ACCELERATION COMPUTATIONS	MAIN	94
	IS EQUAL TO ECG-*,F10.4,7H*BL OR ,F10.4)	MAIN	95
C		MAIN	96
	WRITE(6,23)	MAIN	97
	WRITE(6,47)	MAIN	98
	23 FORMAT(1H ,//)	MAIN	99
	47 FORMAT(" STATION NO.",3X,"DEAD RISE",8X,"EST",8X,"NO",	MAIN	100
	* 10X,"BEAM")	MAIN	101
	WRITE(6,55) ((I,BETA,EST(I),NU(I),BM(I)),I=1,NUM)	MAIN	102
	55 FORMAT(6X,I2,5X,F10.4,4X,F10.4,4X,F10.4,3X,F10.4)	MAIN	103
	WRITE(6,23)	MAIN	104
	WRITE(6,56) (X(I),I=1,6)	MAIN	105
	56 FORMAT(" X VALUES",4X,6(F10.4,2X))	MAIN	106
C	* * * * * CHANGE INPUT FROM DEGREES TO RADIANS	MAIN	107
	X(3) = X(3)*RPD	MAIN	108
	X(6) = X(6)*RPD	MAIN	109
C		MAIN	110
	WAVE = START+RISE	MAIN	111
	NWAVE = 0	MAIN	112
C	* * * * * WRITE OUT COMPUTED ARRAYS	MAIN	113
	WRITE(6,57)M,IT,K,C,PHALF,PI,GRAVITY	MAIN	114
	IF(NPRINT.LT.4) GO TO 62	MAIN	115
	WRITE(6,58) (E(I),I=1,NUM)	MAIN	116
	WRITE(6,59) (N(I),I=1,NUM)	MAIN	117
	WRITE(6,64) (HMAX(I),I=1,NUM)	MAIN	118
	WRITE(6,66) (TEST(I),I=1,NUM)	MAIN	119

62	CONTINUE	MAIN 120
	WRITE(6,28) (KTT(I),DIFF(I),I=1,II)	MAIN 121
28	FORMAT(* KTT,DIFF *,110,2X,F10.4)	MAIN 122
57	FORMAT(4H M= ,F10.4,4H I= ,F10.4,4H K= ,F10.4,4H C= ,F10.4,11H PI=	MAIN 123
	RHO/2= ,F10.4,5H PI= ,F10.4,10H GRAVITY= ,F10.4)	MAIN 124
58	FORMAT (" E(I)",10F10.4)	MAIN 125
59	FORMAT (" N(I)",10F10.4)	MAIN 126
64	FORMAT (" HMAX(I)",10F10.4)	MAIN 127
66	FORMAT (" TEST(I)",10F10.4)	MAIN 128
	IB = 1	MAIN 129
	IPRINT = NPRINT	MAIN 130
	WRITE(4,91)	MAIN 131
C * *	***** WRITE HEADINGS AND CONDITIONS AT TIME = 0.	MAIN 132
91	FORMAT(1H1,2X,"TIME",9X,"XDOT",9X,"ZDOT",9X,"THETA DOT",6X,	MAIN 133
	1HX,9X,1HZ,9X,5H THETA,9X,2HNL,9X,2HFL,	MAIN 134
	4X,8H BW ACCL,4X,7H CG ACCL,//)	MAIN 135
	WRITE(4,92) TIME,(X(I),I=1,6),NL,FL,BWACL,CGACL	MAIN 136
	WRITE(9) TIME,(X(I),I=4,6),BWACL,CGACL	MAIN 137
	KOUNT = KOUNT+1	MAIN 138
	FX(1,IB)=X(5)	MAIN 139
	FX(2,IB)=X(6)	MAIN 140
	IKUTM=(XE-XA)/HMAX+.05	MAIN 141
	IKUTM = (TME-XA)/HMAX + (XE-TME)/HMX + .05	MAIN 142
	FIRST=0.0	MAIN 143
	NEQS=6	MAIN 144
	IKUTS=0	MAIN 145
C		MAIN 146
C	START OF INTEGRATION LOUP	MAIN 147
C		MAIN 148
851	CONTINUE	MAIN 149
	NPRINT = IPRINT	MAIN 150
C * *	***** CHECK PITCH .GT. .5236 RADIAN	MAIN 151
	IF(X(6).GT..5236)GO TO 853	MAIN 152
C * *	***** PERFORM INTEGRATIONS	MAIN 153
	IF(TIME.LT.TM.OR.TME.EQ.XE) GO TO 98	MAIN 154
	IF(IPT.EQ.1) GO TO 98	MAIN 155
	HMIN = HMN	MAIN 156
	HMAX = HMX	MAIN 157
	FIRST = 0.0	MAIN 158
98	CONTINUE	MAIN 159
	CALL KUTMEP(NEQS,TIME,HMAX,X,EPSE,A,HMIN,FIRST)	MAIN 160
	IKUTS=IKUTS+1	MAIN 161
	IF(FIRST.EQ.2)GO TO 861	MAIN 162
	IF(KOUNT.NE.1.AND.KOUNT.NE.41) GO TO 99	MAIN 163
	WRITE(4,91)	MAIN 164
	KOUNT=1	MAIN 165
C * *	***** WRITE OUT TIME INTERVAL RESULTS	MAIN 166
99	WRITE(4,92) TIME,(X(I),I=1,6),NL,FL,BWACL,CGACL	MAIN 167
	WRITE(6,93) T1,T2,T3,T4,T5,T6,T7,T8,BMM,BF	MAIN 168
	WRITE(9) TIME,(X(I),I=4,6),BWACL,CGACL	MAIN 169
	IF(TIME.LT.TM.OR.TME.EQ.XE) GO TO 200	MAIN 170
	IF(IPT.EQ.1) GO TO 200	MAIN 171
	CALL PLUTEP(FX,XA,HMAX,LAMBDA,IB,NWAVE,IPT)	MAIN 172
	IPT = 1	MAIN 173
	IB = 0	MAIN 174
	XA = TIME	MAIN 175
	FIRST = 0.0	MAIN 176
	HMIN = HMN	MAIN 177
	HMAX = HMX	MAIN 178

200	CONTINUE	MAIN 179
	IB=IB+1	MAIN 180
	FX(1,IB)=X(5)	MAIN 181
	FX(2,IB)=X(6)	MAIN 182
93	FORMAT(" ",10E10.4)	MAIN 183
92	FORMAT(1X,11(F10.4,2X))	MAIN 184
100	CONTINUE	MAIN 185
	KOUNT=KOUNT+1	MAIN 186
	IF(NWAVE.GT.0)GO TO 21	MAIN 187
	IF(TIME.GT.WAVE)NWAVE=KOUNT	MAIN 188
21	CONTINUE	MAIN 189
	IF(TIME.LE.XE.AND.IKUTS.LT.IKUTH)GO TO 851	MAIN 190
	WRITE(2,852)	MAIN 191
854	CONTINUE	MAIN 192
852	FORMAT(" END OF KUTMER")	MAIN 193
853	CONTINUE	MAIN 194
	CALL PLUTER(FX,XA,HMAX,LAMBDA,IB,NWAVE,IPT)	MAIN 195
C * *	* * * * * CHECK FOR LAST RUN IF NOT CYCLE BACK TO READ	MAIN 196
C	NEW DATA FOR NEXT RUN	MAIN 197
	IF(END.NE.1)GO TO 8	MAIN 198
	GO TO 999	MAIN 199
C * *	* * * KUTMER ERROR MESSAGES	MAIN 200
861	WRITE(6,862)	MAIN 201
862	FORMAT(" ERROR CRITERION IN KUTMER CAN NOT BE MET")	MAIN 202
	WRITE (6,56) (X(I),I=1,6)	MAIN 203
	WRITE (6,86) TIME	MAIN 204
86	FORMAT (" TIME =",F10.4)	MAIN 205
	IF(END.NE.1)GO TO 8	MAIN 206
	GO TO 853	MAIN 207
999	CONTINUE	MAIN 208
	END FILE 9	MAIN 209
	END	MAIN 210
	SUBROUTINE PLUT2(F,FMIN,FMAX,NVAR,NFUN,N1,N,X0,DELX)	PLOT2 2
C		PLOT2 3
C	PLUT FIRST N POINTS OF UP TO 26 FUNCTIONS F(X)	PLOT2 4
C	F(I,J) CONTAINS THE VALUE FOR THE JTH POINT OF THE ITH FUNCTION	PLOT2 5
C	FMIN(I) AND FMAX(I) CONTAIN THE MIN AND MAX ORDINATE VALUES FOR	PLOT2 6
C	THE ITH FUNCTION.	PLOT2 7
C	NVAR(I) AN ARRAY OF TITLES FOR THE VARIOUS FUNCTIONS	PLOT2 8
C	TO BE PLOTTED AGAINST THE ABSCISSA	PLOT2 9
C	NFUN NUMBER OF FUNCTIONS TO BE PLOTTED - DIMENSION OF	PLOT2 10
C	NVAR, FMIN, FMAX	PLOT2 11
C	N1 USED ONLY IN F(N1,1) AS PASSED DIMENSION	PLOT2 12
C	N NUMBER OF POINTS IN A SINGLE PLOT FRAME	PLOT2 13
C	X0 FIRST ABSCISSA VALUE	PLOT2 14
C	DELX ABSCISSA INCREMENT	PLOT2 15
C		PLOT2 16
	DIMENSION NSTEP(26),F(N1,N),FMIN(NFUN),FMAX(NFUN),VLAST(26),	PLOT2 17
1	VFI,ST(26),HEAD(6),STEP(26)	PLOT2 18
	INTEGER CH(26),NVAR(NFUN),DOT,ASTER,PLUS,BLANK	PLOT2 19
	INTEGER C	PLOT2 20
	INTEGER A(101)	PLOT2 21
C		PLOT2 22
	DATA BLANK,DOT,ASTER,PLUS/1H,1H*,1H*/	PLOT2 23
	DATA CH(1),CH(2),CH(3),CH(4),CH(5),CH(6),CH(7),CH(8),CH(9),CH(10)	PLOT2 24
2	/ 1HA, 1HB, 1HC, 1HD, 1HE, 1HF, 1HG, 1HH, 1HI, 1HJ /	PLOT2 25
	DATA CH(11),CH(12),CH(13),CH(14),CH(15),CH(16),CH(17),CH(18)	PLOT2 26
2	/ 1HK, 1HL, 1HM, 1HN, 1HO, 1HP, 1HQ, 1HR/	PLOT2 27
	DATA CH(19),CH(20),CH(21),CH(22),CH(23),CH(24),CH(25),CH(26)	PLOT2 28

2	/ 1HS , 1HT , 1HU , 1HV , 1HW , 1HX , 1HY , 1HZ /	PLOT2 29
C	IF(NFUN.LE.0.OR.N.LE.0) RETURN	PLOT2 30
C	PRINT HEADINGS.	PLOT2 31
	WRITE(6,46)	PLOT2 32
46	FORMAT (///)	PLOT2 33
	DO 40 I=1,NFUN	PLOT2 34
30	TENM=ABS(FMAX(I)-FMIN(I))	PLOT2 35
	EXP=1.	PLOT2 36
	IF (TENM.EQ.0.) GO TO 2	PLOT2 37
C	BRING TENM TO A VALUE BETWEEN 1 AND 10	PLOT2 38
	IF(TENM.LT.1.) GO TO 1	PLOT2 39
3	IF(TENM.LT.10.) GO TO 2	PLOT2 40
	EXP=EXP*10.	PLOT2 41
	TENM=TENM*.1	PLOT2 42
	GO TO 3	PLOT2 43
1	EXP=EXP*.1	PLOT2 44
	TENM=TENM*10.	PLOT2 45
	IF(TENM.GT.1.) GO TO 2	PLOT2 46
	GO TO 1	PLOT2 47
C	SET UP VALUE BETWEEN GRID LINES, RSTEP.	PLOT2 48
2	PSTEP=5.	PLOT2 49
	IF(TENM.GE.5.)PSTEP=10.	PLOT2 50
	IF(TENM.LT.2.)PSTEP=2.	PLOT2 51
5	RSTEP(I)=PSTEP*EXP*.1	PLOT2 52
C	COMPUTE VALUE OF STARTING LINE, VFIRST.	PLOT2 53
	FIRST=FMIN(I)/RSTEP(I)	PLOT2 54
	IF(FMIN(I).LT.0.)FIRST=FIRST-1.	PLOT2 55
	FIRST=AINT(FIRST)	PLOT2 56
	VFIRST(I)=FIRST*RSTEP(I)	PLOT2 57
C	CHECK END LINE VALUE,VLAST.	PLOT2 58
	VLAST(I)=VFIRST(I)+10.*RSTEP(I)	PLOT2 59
	IF(VLAST(I).GT.FMAX(I))GO TO 4	PLOT2 60
C	IF GRAPH IS TOO SMALL TAKE NEXT LARGER STEP.	PLOT2 61
	AA=PSTEP	PLOT2 62
	IF(AA.LT.5.)PSTEP=5.	PLOT2 63
	IF(AA.EQ.5.)PSTEP=10.	PLOT2 64
	IF(AA.LT.10.) GO TO 5	PLOT2 65
	PSTEP=2.	PLOT2 66
	EXP=10.*EXP	PLOT2 67
	GO TO 5	PLOT2 68
C	COMPUTE VALUE BETWEEN POINTS,STEP.	PLOT2 69
4	STEP(I)=RSTEP(I)*.1	PLOT2 70
	RK=0.	PLOT2 71
	DO 6 KK=1,6	PLOT2 72
	HEAD(KK)=VFIRST(I)+2.*RK*RSTEP(I)	PLOT2 73
6	RK=RK+1.	PLOT2 74
40	WRITE(6,45) CH(I), NVAR(I), (HEAD(KK),KK=1,6)	PLOT2 75
45	FORMAT(1X,A1,3H = ,A10,5X,1PE12.4,5(8X,1PE12.4))	PLOT2 76
	DO 50 J=1,101	PLOT2 77
	A(J)=BLANK	PLOT2 78
	IF(MOD(J,10).EQ.1) A(J)=DOT	PLOT2 79
50	CONTINUE	PLOT2 80
	WRITE(6,55) A,A	PLOT2 81
55	FORMAT (25X,101A1/15X,4HTIME,6X,101A1)	PLOT2 82
C	PLUT EACH POINT	PLOT2 83
	DO 100 J=1,N	PLOT2 84
	B=X0+FLUAT(J-1)*DELX	PLOT2 85
	DO 70 K=1,101	PLOT2 86
		PLOT2 87

A(K)=BLANK	PLOT2 88
IF(MOD(K,10).EQ.1) A(K)=DOT	PLOT2 89
IF(MOD(J,5).EQ.1) A(K)=DOT	PLOT2 90
70 CONTINUE	PLOT2 91
DO 80 I=1,NFUN	PLOT2 92
LOC=((F(I,J)-VFIRST(I))/STEP(I)+1.5)	PLOT2 93
C=A(LOC)	PLOT2 94
A(LOC)=CH(I)	PLOT2 95
IF(C.NE.BLANK.AND.C.NE.DOT) A(LOC)=ASTER	PLOT2 96
80 CONTINUE	PLOT2 97
IF(MOD(J,10).EQ.1)GO TO 95	PLOT2 98
WRITE(6,85) A	PLOT2 99
85 FORMAT (25X,101A1)	PLOT2100
GO TO 100	PLOT2101
95 WRITE(6,15)B,A	PLOT2102
15 FORMAT (12X,1PE12.4,1X,101A1)	PLOT2103
100 CONTINUE	PLOT2104
RETURN	PLOT2105
END	PLOT2106
SUBROUTINE KUTMER(ND,T,H,Y0,EPSE,A,HCX,FIRST)	KUTMER 2
DIMENSION Y0(6),Y1(6),Y2(6),F0(6),F1(6),F2(6),EPSE(6),A(6)	KUTMER 3
COMMON/UUT,NPRINT,NPLOT,END	KUTMER 4
COMMON /ACCEL / XACCL,BWACL,CGACL,BL	KUTMER 5
DATA NAM1,NAM2 /2HY1,2HY2 /	KUTMER 6
C	KUTMER 7
ND = NUMBER OF EQUATIONS, NO. OF COMPONENTS OF Y0	KUTMER 8
C T = INDEPENDENT VARIABLE	KUTMER 9
C H = INCREMENT FOR WHICH SOLUTION IS TO BE RETURNED + OR -	KUTMER10
C Y0 = THE VECTOR OF DEPENDENT VARIABLES. ENTER WITH INITIAL	KUTMER11
VALUES AT T AND RETURN WITH VALUES AT T+H	KUTMER12
C EPSE = RELATIVE ERROR CRITERION FOR COMPONENTS OF Y0 ,GT ABS(A)	KUTMER13
C A = ABSOLUTE ERROR SRITERION FOR COMPONENTS OF Y0 ,LT. ABS(A)	KUTMER14
C NOTE-- EPSE AND A MUST BE SPECIFIED FOR EACH COMPONENT OF THE SYSTEM	KUTMER15
C HCX = THE SMALLEST STEP SIZE USED IN THE INTEGRATION	KUTMER16
C FIRST SHOULD BE 0 WHEN KUTMER IS ENTERED FOR THE FIRST TIME	KUTMER17
C AFTER THAT FIRST IS 1 IF KUTMER IS ENTERED WITH THE SAME H OR	KUTMER18
C IF IT IS ENTERED WITH A CHANGED H	KUTMER19
C IF FIRST IS 2 THE ERROR SRITERIA CANNOT BE MEET AND THE STEP SIZE	KUTMER20
C REDUCED TO H/128.	KUTMER21
C	KUTMER22
IF (FIRST) 20,10,20	KUTMER23
C - - - - - FIRST ENTRY	KUTMER24
10 HC = H	KUTMER25
IPLOC = 1	KUTMER26
FIRST = 1.	KUTMER27
C - - - - - OTHER ENTRY	KUTMER28
20 LOC = 0	KUTMER29
HCX = HC	KUTMER30
IF (HC.NE.0.) GO TO 30	KUTMER31
WRITE(6,800)	KUTMER32
800 FORMAT(5X,45HKUTMER ENTERED WITH ZERO INTEGRATION INTERVAL)	KUTMER33
FIRST = 2.	KUTMER34
RETURN	KUTMER35
C - - - - - 5 CALLS TO DAUX	KUTMER36
30 CALL DAUX(T,Y0,F0)	KUTMER37
IF(NPRINT.EQ.5)WRITE(6,400)Y0,T,F0	KUTMER38
400 FORMAT(6(2X,F10.4),4HTIME,2X,F10.4)	KUTMER39
IF(NPRINT.EQ.5)WRITE(6,400)HC	KUTMER40
39 DO 40 I=1,ND	KUTMER41

40	Y1(I) = Y0(I)*(HC/3.)*F0(I)	KUTHER42
	IF(NPRINT.EQ.5)WRITE(6,400)Y1,T	KUTHER43
C		KUTHER44
	CALL DAUX(T+HC/3.,Y1,F1)	KUTHER45
	IF(NPRINT.EQ.5)WRITE(6,400)F1,T	KUTHER46
	DO 50 I=1,ND	KUTHER47
50	Y1(I) = Y0(I)*(HC/6.)*F0(I)+(HC/6.)*F1(I)	KUTHER48
	IF(NPRINT.EQ.5)WRITE(6,400)Y1,T	KUTHER49
C		KUTHER50
	CALL DAUX(T+HC/3.,Y1,F1)	KUTHER51
	IF(NPRINT.EQ.5)WRITE(6,400)F1,T	KUTHER52
	DO 60 I=1,ND	KUTHER53
60	Y1(I) = Y0(I)*(HC/8.)*F0(I)+.375*HC*F1(I)	KUTHER54
	IF(NPRINT.EQ.5)WRITE(6,400)Y1,T	KUTHER55
C		KUTHER56
	CALL DAUX(T+HC/2.,Y1,F2)	KUTHER57
	IF(NPRINT.EQ.5)WRITE(6,400)F2,T	KUTHER58
	DO 70 I=1,ND	KUTHER59
70	Y1(I) = Y0(I)*(HC/2.)*F0(I)-1.5*HC*F1(I)+2.*HC*F2(I)	KUTHER60
	IF(NPRINT.EQ.5)WRITE(6,400)Y1,T	KUTHER61
C		KUTHER62
	CALL DAUX(T+HC,Y1,F1)	KUTHER63
	IF(NPRINT.EQ.5)WRITE(6,400)F1,T	KUTHER64
	DO 80 I=1,ND	KUTHER65
80	Y2(I) = Y0(I)+HC/6.*F0(I)+(2./3.)*HC*F2(I)+(HC/6.)*F1(I)	KUTHER66
	IF(NPRINT.EQ.5)WRITE(6,400)Y2,T	KUTHER67
	INC = 0	KUTHER68
C	----- CHECK ERROR CRITERIA	KUTHER69
	DO 110 I=1,ND	KUTHER70
	ZZZ = ABS(Y1(I))-A(I)	KUTHER71
	IF (ZZZ) 85,87,87	KUTHER72
C	----- ABSOLUTE ERROR	KUTHER73
85	ERROR = ABS(.2*(Y1(I)-Y2(I)))	KUTHER74
	IF (ERROR-A(I)) 100,100,90	KUTHER75
C	----- RELATIVE ERROR	KUTHER76
87	ERROR = ABS(.2-.2*Y2(I)/Y1(I))	KUTHER77
	IF (ERROR-EPSE(I)) 100,100,90	KUTHER78
C	----- SINCE ERROR .GT. ERROR CRITERIA CHECK IF HC.GT.H/KUTHER79	KUTHER79
C	----- IF YES THEN HALVE INTERVAL. OTHERWISE STOP.	KUTHER80
90	X = 128.*ABS(HC)-ABS(H)	KUTHER81
	IF(X) 91,95,95	KUTHER82
C	----- ERROR TOO LARGE	KUTHER83
91	WRITE(6,92)I,T,ERROR,HC	KUTHER84
92	FORMAT(/18H FOR EQUATION NO. 12,27H, THE RELATIVE ERROR AT T = ,	KUTHER85
	E15.8, 4H IS ,E15.8,13H STEP SIZE = ,E15.8)	KUTHER86
	FIRST = 2.	KUTHER87
	RETURN	KUTHER88
C	----- HALVE INTERVAL	KUTHER89
95	HC = HC/2.	KUTHER90
	IPLOC = 2*IPLOC	KUTHER91
	LOC = 2*LOC	KUTHER92
	HXC = HC	KUTHER93
	WRITE(2,71)T,I,ERROR,HC	KUTHER94
710	FORMAT(/8H TIME = ,F10.3,5X,26HHALVE INTERVAL. EQUATION ,I3,	KUTHER95
	13H HAS ERROR = ,E16.8,6X,17H STEP SIZE NOW = ,E15.8)	KUTHER96
	WRITE(2,720) NAM2,(Y2(J),J=1,ND)	KUTHER97
	WRITE(2,720) NAM1,(Y1(J),J=1,ND)	KUTHER98
720	FORMAT(2X,A2 / 3(10E13.5/))	KUTHER99
	GO TO 30	KUTHE100

C	----	TEST IF INTERVAL LENGTH CAN BE DOUBLED	KUTME101
	100	IF (ERROR*64.-EPSE(I)) 110,110,101	KUTME102
	101	INC = 1	KUTME103
	110	CONTINUE	KUTME104
C	----	UPDATE T AND SOLUTION	KUTME105
	111	T = T+HC	KUTME106
		DO 112 I=1,ND	KUTME107
	112	Y0(I) = Y2(I)	KUTME108
C	----	GET SOLUTION IN NEXT INTERVAL	KUTME109
		L0C = LUC+1	KUTME110
		IF (L0C-IPLOC) 120,210,210	KUTME111
	120	IF (INC) 210,130,210	KUTME112
	130	IF (L0C-(L0C/2)*2) 210,140,210	KUTME113
	140	IF (IPLOC-1) 210,210,200	KUTME114
C	----	DOUBLE INTERVAL LENGTH	KUTME115
	200	HC = 2.*HC	KUTME116
		L0C = LUC /2	KUTME117
		IPLOC = IPLOC/2	KUTME118
	210	IF (IPLOC-L0C) 30,329,30	KUTME119
	329	BWACL = F0(2)-XACCL*F0(3)	KUTME120
		CGACL = F0(2)	KUTME121
		RETURN	KUTME122
		END	KUTME123
		END	KUTME124
		SUBROUTINE DAUX(TIME,X,RHS)	DAUX 2
C			DAUX 3
	TIME	TIME AT WHICH SYSTEM IS TO BE EVALUATED	DAUX 4
C	X	STATE VECTOR	DAUX 5
C	RHS	THE RIGHT HAND SIDE OF THE EQUATION S = F A	DAUX 6
C			DAUX 7
	REAL KAR		DAUX 8
	REAL IA,IT,M,K,MA,MASS,NCG,NL,N,MMA		DAUX 9
	INTEGER ENO,PTIME		DAUX 10
	DIMENSION X(6),RHS(6),F(3,1),A(3,3),INDEX(3,3),		DAUX 11
	• R(120),V(120),D(120)		DAUX 12
C			DAUX 13
	COMMON /SHIP/ MASS,CINT,QA,CE,CE2,CE3,DMU,EDMU,E2DMU,E3DMU,BF,BMH,		DAUX 14
	• NL,FL,IA,E(120)		DAUX 15
	COMMON /CONST/ NCG,ECG,PI,DPR,RPD,GRAVITY,RHO,K,NUM,MA(120),CD,TA,		DAUX 16
	• B(120),BETA,HW(120),TZ,DRAG,W,XD,T,XP,M,IT,		DAUX 17
	• DELTAS,TX,EST(120),C,RO,KAR,MMA(1:0),TEST(120),		DAUX 18
	• N(120),PHALF		DAUX 19
	COMMON /IN/ BM(120),B1(120),VELIN		DAUX 20
	COMMON/UUT/NPRINT,NPLOT,END		DAUX 21
	COMMON /SEAWAVE/ START,RISE,RAMP		DAUX 22
	COMMON /WAVE/ R,PT(120),ZMA,ZWMA,EMAS,ZZWMA,ZWEMA,ZZWMA,E2MAZ,		DAUX 23
	• ZWDOT(120)		DAUX 24
C			DAUX 25
	RAMP = RMP(TIME,START,RISE)		DAUX 26
	PIH = PI/2,		DAUX 27
	CT = C*TIME		DAUX 28
	CX6 = COS(X(6))		DAUX 29
	SX6 = SIN(X(6))		DAUX 30
C	*****SET VALUES OF MA AND B		DAUX 31
	DO 75 I=1,NUM		DAUX 32
	PT(I) = (X(4)*E(I)*CX6+N(I)*SX6*CT)*K		DAUX 33
	R(I) = RU*COS(PT(I))*RAMP		DAUX 34
C	***** COMPUTE HW SUBMERGENCE OF A POINT AND R THE WAVE		DAUX 35
C	HW(I) IS IN THE FIXED COORUINATE SYSTEM		DAUX 36

	HW(I) = X(5)-E(I)*SX6+N(I)*CX6-R(I)	DAUX 37
	IF (HW(I).GT.0) GO TO 65	DAUX 38
C	CRAFT IS NOT SUBMERGED	DAUX 39
	MA(I) = 0.	DAUX 40
	B1(I)=0.	DAUX 41
	B(I) = 0.	DAUX 42
	GO TO 75	DAUX 43
65	V(I) = -RO*K*SIN(PT(I))*RAMP	DAUX 44
	D(I) = HW(I)/(CX6-V(I)*SX6)	DAUX 45
C	D(I) IS IN THE BODY AXIS SYSTEM AND IS THE SUBMERGENCE	DAUX 46
	IF (D(I).GE.TEST(I)) GO TO 70	DAUX 47
C	CRAFT IS PARTLY SUBMERGED	DAUX 48
	B(I) = D(I)*(1./TA)*PIH	DAUX 49
	B1(I) = D(I)*(1./TA)*PIH	DAUX 50
	MA(I) = KAR*PHALF*B(I)*B(I)	DAUX 51
	GO TO 75	DAUX 52
C	CHINE IS IMMERSED	DAUX 53
C	B1 ARRAY IS USED FOR THE INTEGRALS OVER THE PORTION	DAUX 54
C	OF THE HULL FOR WHICH THE CHINE IS NOT IMMERSD	DAUX 55
	70 MA(I)=MMAX(I)	DAUX 56
	B(I)=BM(I)	DAUX 57
	B1(I)=0.	DAUX 58
75	CONTINUE	DAUX 59
	IF (NPRINT.LT.4) GO TO 85	DAUX 60
	WRITE(6,74) TIME	DAUX 61
74	FORMAT(" TIME = ",F10.4)	DAUX 62
	WRITE(6,76) (X(I),I=1,6)	DAUX 63
	WRITE(6,77) (R(I),I=1,NUM)	DAUX 64
	WRITE (6,78) (HW(I),I=1,NUM)	DAUX 65
	WRITE (6,79) (B(I),I=1,NUM)	DAUX 66
	WRITE (6,80) (V(I),I=1,NUM)	DAUX 67
	WRITE (6,81) (D(I),I=1,NUM)	DAUX 68
	WRITE(6,82) (MA(I),I=1,NUM)	DAUX 69
76	FORMAT(" X(I) ",6(2X,E12.6))	DAUX 70
77	FORMAT (" R(I)",10F10.4)	DAUX 71
78	FORMAT (" HW(I)",10F10.4)	DAUX 72
79	FORMAT (" B(I)",10F10.4)	DAUX 73
80	FORMAT (" V(I)",10F10.4)	DAUX 74
81	FORMAT (" D(I)",10F10.4)	DAUX 75
82	FORMAT(" MA(I) ",10F10.4)	DAUX 76
85	CONTINUE	DAUX 77
C		DAUX 78
C	* * * * * COMPUTES NL AND FL AND THE ASSOCIATED INTERGALS	DAUX 79
	CALL FUNCT(X)	DAUX 80
C		DAUX 81
	IF (NPRINT.LT.4) GO TO 17	DAUX 82
	WRITE(6,15) TX,FL,DRAG,TZ,W,NL,XD,T,XP	DAUX 83
15	FORMAT(" ",10E12.6)	DAUX 84
17	CONTINUE	DAUX 85
C	* * * * * COMPUTE THE F VECTOR	DAUX 86
	F(1,1) = TX+FL*SX6-DRAG*CX6	DAUX 87
	F(1,1)=0.0	DAUX 88
	F(2,1) = TZ+FL*CX6+DRAG*SX6+W	DAUX 89
	F(3,1)=NL-DRAG*XD+T*XP	DAUX 90
	IF (NPRINT.LT.3) GO TO 18	DAUX 91
	WRITE(6,10) (F(I,1),I=1,3)	DAUX 92
18	CONTINUE	DAUX 93
C	* * * * * COMPUTE THE A MATRIX	DAUX 94
	A(1,1) = M*MASS*SX6*SX6	DAUX 95

A(1,2) = MASS* $SX6 \times CX6$	DAUX 96
A(1,3) = - $QA \times SX6$	DAUX 97
A(1,2) = 0.	DAUX 98
A(1,3) = 0.	DAUX 99
A(2,1)=A(1,2)	DAUX 100
A(2,2) = M+MASS* $CX6 \times CX6$	DAUX 101
A(2,3) = - $QA \times CX6$	DAUX 102
A(3,1)=A(1,3)	DAUX 103
A(3,2)=A(2,3)	DAUX 104
A(3,3)=IT+JA	DAUX 105
IF(NPRINT,LT,3)GO TO 25	DAUX 106
WRITE(6,12) (A(I,1),I=1,3)	DAUX 107
WRITE(6,13) (A(I,2),I=1,3)	DAUX 108
WRITE(6,14) (A(I,3),I=1,3)	DAUX 109
C * * * * * [INVERT THE A MATRIX	DAUX 110
25 CALL MATINS(A,3,3,F,1,1,DETERM,ID,INDEX)	DAUX 111
IF(ID,EQ,2)WRITE(6,26)	DAUX 112
26 FORMAT(" MATRIX IS SINGULAR ")	DAUX 113
C*****A ON RETURN WILL CONTAIN THE INVERSE MATRIX	DAUX 114
C ID=2 MATRIX IS SINGULAR	DAUX 115
C =1 INVERSE WAS FOUND	DAUX 116
C	DAUX 117
C * * * * * COMPUTE THE RIGHT HAND SIDE	DAUX 118
RHS(1) = F(1,1)	DAUX 119
RHS(2) = F(2,1)	DAUX 120
RHS(3) = F(3,1)	DAUX 121
RHS(1) = 0.0	DAUX 122
RHS(4) = X(1)	DAUX 123
RHS(5) = X(2)	DAUX 124
RHS(6) = X(3)	DAUX 125
10 FORMAT(" F(I,1) ",3(2X,E12.4))	DAUX 126
12 FORMAT(" A(I,1) ",3(2X,E12.4))	DAUX 127
13 FORMAT(" A(I,2) ",3(2X,E12.4))	DAUX 128
14 FORMAT(" A(I,3) ",3(2X,E12.4))	DAUX 129
30 IF(NPRINT,LT,2) GO TO 40	DAUX 130
WRITE(6,12) (A(I,1),I=1,3)	DAUX 131
WRITE(6,13) (A(I,2),I=1,3)	DAUX 132
WRITE(6,14) (A(I,3),I=1,3)	DAUX 133
WRITE(6,35) (RHS(I),I=1,6)	DAUX 134
35 FORMAT(" RHS(I) ",6(2X,E12.6))	DAUX 135
40 CONTINUE	DAUX 136
RETURN	DAUX 137
END	DAUX 138
SUBROUTINE FUNCT(X)	FUNCT 2
REAL KAR	FUNCT 3
REAL IA,IAA,IPART,K,KPI,MA,MASS,NL,NCG,IT,M,MMAX,N	FUNCT 4
INTEGER END	FUNCT 5
DIMENSION IPART(120),C1(120),C2(120),	FUNCT 6
• D1(120),D2(120),D3(120),D4(120),D5(120),D6(120),	FUNCT 7
• QPART(120),Z1(120),Z2(120),Z3(120),Z4(120),Z5(120),	FUNCT 8
• Z6(120),Z7(120)	FUNCT 9
• ,X(6),VMMA(120)	FUNCT 10
C	FUNCT 11
COMMON /SHIP/ MASS,CINT,QA,CE,CE2,CE3,DMU,EDMU,E2DMU,E3DMU,BF,BMM,	FUNCT 12
• NL,FL,IA,E(120)	FUNCT 13
COMMON /CONST/ NCG,ECG,P1,DPR,RPD,GRAVITY,RHO,K,NUM,MA(120),CD,TA,	FUNCT 14
• B(120),BETA,HW(120),TZ,DRAG,W,XD,T,XP,M,IT,	FUNCT 15
• DELTAS,TX,EST(120),C,RO,KAR,MMA(1 0),TEST(120),	FUNCT 16
• N(120),PHALF	FUNCT 17

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COMMON /IN/ BM(120),B1(120),VELIN                                FUNCT 18
COMMON/OUT,NPRINT,NPLOT,END                                        FUNCT 19
COMMON /WAVE/ R(120),PT(120),ZMA,ZWMA,EMAS,ZZWMA,ZWEMA,ZZWMA,EZMAZFUNCT 20
* ,ZWDOT(120)                                                    FUNCT 21
COMMON /INTER/ I1,KTT(10),DIFF(10)                                FUNCT 22
COMMON /SEAWAVE/ START,RISE,RAMP                                  FUNCT 23
COMMON /TEST/ VMA                                                  FUNCT 24
C * * * * * INITIALIZE INTEGRAL SUMS                               FUNCT 25
MASS = 0.0                                                         FUNCT 26
QA = 0.0                                                           FUNCT 27
IA = 0.0                                                           FUNCT 28
CE = 0.0                                                           FUNCT 29
CE2 = 0.0                                                         FUNCT 30
DMU = 0.0                                                         FUNCT 31
EDMU=0.0                                                           FUNCT 32
E2DMU = 0.0                                                       FUNCT 33
E3DMU = 0.0                                                       FUNCT 34
BF = 0.0                                                           FUNCT 35
BMM = 0.0                                                         FUNCT 36
ZMA = 0.0                                                         FUNCT 37
ZWMA = 0.0                                                         FUNCT 38
EMAS = 0.0                                                         FUNCT 39
ZZWMA = 0.0                                                       FUNCT 40
ZWEMA = 0.0                                                       FUNCT 41
ZZWMA = 0.0                                                       FUNCT 42
EZMAZ = 0.0                                                       FUNCT 43
VPART = X(1)*SIN(X(6))+X(2)*COS(X(6))                             FUNCT 44
SX6 = SIN(X(6))                                                    FUNCT 45
CX6 = COS(X(6))                                                    FUNCT 46
W0 = K*C                                                           FUNCT 47
C * * * * * SET UP THE FUNCTIONS FOR THE INTEGRALS (PAGE 4 OF NOFUNCT 48
DO 90 I=1,NUM                                                       FUNCT 49
IPART(I)=E(I)*E(I)*MA(I)                                           FUNCT 50
QPART(I)=E(I)*MA(I)                                                FUNCT 51
ZWDOT(I) = -RU*W0*SIN(PT(I))*RAMP                                  FUNCT 52
U = X(1)*CX6-X(2)*SX6+ZWDOT(I)*SX6                                FUNCT 53
VEL = VPART-X(3)*E(I)-ZWDOT(I)*CX6                                FUNCT 54
Z1(I) = MA(I)*ZWDOT(I)                                             FUNCT 55
Z2(I) = -MA(I)*COS(PT(I))*RAMP                                     FUNCT 56
Z3(I) = E(I)*Z2(I)                                                 FUNCT 57
Z4(I) = E(I)*Z1(I)                                                 FUNCT 58
Z5(I) = U*Z2(I)                                                    FUNCT 59
Z6(I) = E(I)*Z5(I)                                                 FUNCT 60
Z7(I) = MA(I)*VEL*U                                                FUNCT 61
IF (VEL.LE.0.) GO TO 60                                           FUNCT 62
IF (R1(I).LE.0.0) GO TO 50                                         FUNCT 63
DRDT = ZWDOT(I)*(X(1)+C*X(3)*(N(I)*CX6-E(I)*SX6))/C              FUNCT 64
D1(I) = VEL*B1(I)*(X(2)-X(3)*(CX6*E(I)+SX6*N(I)) -DRDT)         FUNCT 65
GO TO 51                                                            FUNCT 66
50 D1(I) = 0.                                                       FUNCT 67
51 CONTINUE                                                         FUNCT 68
D2(I) = E(I)*D1(I)                                                 FUNCT 69
C1(I) = VEL*VEL*B(I)                                               FUNCT 70
C2(I) = E(I)*C1(I)                                                 FUNCT 71
GO TO 61                                                            FUNCT 72
60 D1(I) = 0.                                                       FUNCT 73
D2(I) = 0.                                                         FUNCT 74
C1(I) = 0.                                                         FUNCT 75
C2(I) = 0.                                                         FUNCT 76

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61	CONTINUE	FUNCT 77
	D3(I) = Z2(I)*VEL	FUNCT 78
	D4(I) = E(I)*D3(I)	FUNCT 79
	PIH = PI/2.	FUNCT 80
	.D5(I) = B(I)*(HW(I)-B(I)*TA/2.)	FUNCT 81
66	D6(I) = D5(I)*E(I)*.5	FUNCT 82
90	CONTINUE	FUNCT 83
	RHOG=RHU*GRAVITY	FUNCT 84
C *	* * * * * SET UP THE FUNCTIONS FOR THE INTEGRALS (PAGE 5 OF NOTES)	FUNCT 85
	PIH = PI/2.	FUNCT 86
	KPI = KAR*PI	FUNCT 87
C	EVALUATE INTEGRALS USING TRAP METHOD	FUNCT 88
	I = 1	FUNCT 89
	INDEX = 1	FUNCT 90
91	CALL TRAP(MA(INDEX),DIFF(I),KTT(I),TMASS)	FUNCT 91
	CALL TRAP(OPART(INDEX),DIFF(I),KTT(I),QA1)	FUNCT 92
	CALL TRAP(C1(INDEX),DIFF(I),KTT(I),CEA)	FUNCT 93
	CALL TRAP(C2(INDEX),DIFF(I),KTT(I),CE2A)	FUNCT 94
	CALL TRAP(IPART(INDEX),DIFF(I),KTT(I),IAA)	FUNCT 95
	CALL TRAP(D1(INDEX),DIFF(I),KTT(I),DMUA)	FUNCT 96
	CALL TRAP(D2(INDEX),DIFF(I),KTT(I),EDMUA)	FUNCT 97
	CALL TRAP(D3(INDEX),DIFF(I),KTT(I),E2DMUA)	FUNCT 98
	CALL TRAP(D4(INDEX),DIFF(I),KTT(I),E3DMUA)	FUNCT 99
	CALL TRAP(D5(INDEX),DIFF(I),KTT(I),BFA)	FUNCT100
	CALL TRAP(D6(INDEX),DIFF(I),KTT(I),BMAA)	FUNCT101
	CALL TRAP(Z1(INDEX),DIFF(I),KTT(I),ZMAA)	FUNCT102
	CALL TRAP(Z2(INDEX),DIFF(I),KTT(I),ZWMAA)	FUNCT103
	CALL TRAP(Z3(INDEX),DIFF(I),KTT(I),EMASA)	FUNCT104
	CALL TRAP(Z4(INDEX),DIFF(I),KTT(I),ZZWMAA)	FUNCT105
	CALL TRAP(Z5(INDEX),DIFF(I),KTT(I),ZWEMAA)	FUNCT107
	CALL TRAP(Z6(INDEX),DIFF(I),KTT(I),Z2WMAA)	FUNCT108
	CALL TRAP(Z7(INDEX),DIFF(I),KTT(I),E2MAZA)	FUNCT109
C		FUNCT110
93	CONTINUE	FUNCT111
	MASS = MASS + TMASS	FUNCT112
	QA = QA + QA1	FUNCT113
	IA = IA + IAA	FUNCT114
	CE = CE + CEA	FUNCT115
	CE2 = CE2 + CE2A	FUNCT116
	DMU = DMU + DMUA	FUNCT117
	EDMU = EDMU + EDMUA	FUNCT118
	E2DMU = E2DMU + E2DMUA	FUNCT119
	E3DMU = E3DMU + E3DMUA	FUNCT120
	BF = BF + OHUG*BFA	FUNCT121
	BMM = BMM + RHOG*BMAA	FUNCT122
	ZMA = ZMA+ZMAA	FUNCT123
	ZWMA = ZWMA+ZWMAA	FUNCT124
	EMAS = EMAS+EMASA	FUNCT125
	ZZWMA = ZZWMA+ZZWMAA	FUNCT126
	ZWEMA = ZWEMA+ZWEMAA	FUNCT127
	Z2WMA = Z2WMA+Z2WMAA	FUNCT128
	E2MAZ = E2MAZ+E2MAZA	FUNCT129
94	CONTINUE	FUNCT130
	IF (I,GE,II)GO TO 92	FUNCT131
	INDEX = INDEX+KTT(I)-1	FUNCT132
	I = I+1	FUNCT133
	GO TO 91	FUNCT134
92	CONTINUE	FUNCT135
C		

C * * * * * CALL COMPUT TO FIND THE VALUE OF NL AND FL USING	FUNCT136
C THE VALUES OF THE ABOVE INTEGRALS	FUNCT137
CALL COMPUT(X)	FUNCT138
C	FUNCT139
IF(NPRINT,LT,3) GO TO 111	FUNCT140
IF(NPRINT,EQ,3) GO TO 108	FUNCT141
IF(NPRINT,EQ,4) GO TO 108	FUNCT142
WRITE(6,97) (IPART(I),I=1,NUM)	FUNCT143
WRITE(6,98) (OPART(I),I=1,NUM)	FUNCT144
WRITE(6,99) (C1(I),I=1,NUM)	FUNCT145
WRITE(6,100) (C2(I),I=1,NUM)	FUNCT146
WRITE(6,101) (C3(I),I=1,NUM)	FUNCT147
WRITE(6,102) (D1(I),I=1,NUM)	FUNCT148
WRITE(6,103) (D2(I),I=1,NUM)	FUNCT149
WRITE(6,104) (D3(I),I=1,NUM)	FUNCT150
WRITE(6,105) (D4(I),I=1,NUM)	FUNCT151
WRITE(6,106) (D5(I),I=1,NUM)	FUNCT152
WRITE(6,112) (D6(I),I=1,NUM)	FUNCT153
WRITE(6,113) (Z1(I),I=1,NUM)	FUNCT154
WRITE(6,114) (Z2(I),I=1,NUM)	FUNCT155
WRITE(6,115) (Z3(I),I=1,NUM)	FUNCT156
WRITE(6,116) (Z4(I),I=1,NUM)	FUNCT157
WRITE(6,118) (Z5(I),I=1,NUM)	FUNCT158
WRITE(6,119) (Z6(I),I=1,NUM)	FUNCT159
WRITE(6,120) (Z7(I),I=1,NUM)	FUNCT160
WRITE(6,107) KPI,RHOG,PIH	FUNCT161
108 WRITE(6,109) MASS,CINT,QA,CE,CE2,CE3	FUNCT162
WRITE(6,121) IA	FUNCT163
121 FORMAT(* IA *,E10,4)	FUNCT164
WRITE(6,110) DMU,EDMU,E2DMU,E3DMU,BF,BMM	FUNCT165
WRITE(6,117) ZMA,ZWMA,EMAS,ZZWMA,ZWEMA,ZZWMA,E2MAZ	FUNCT166
C * * * * * FORMATS * * * * * * * * * * * * * * *	FUNCT167
96 FORMAT(" CPART(I)",10(2X,E10,4))	FUNCT168
97 FORMAT(" IPART(I)",10(2X,E10,4))	FUNCT169
98 FORMAT(" OPART(I)",10(2X,E10,4))	FUNCT170
99 FORMAT(" C1 ",10(2X,E10,4))	FUNCT171
100 FORMAT(" C2 ",10(2X,E10,4))	FUNCT172
101 FORMAT(" C3 ",10(2X,E10,4))	FUNCT173
102 FORMAT(" D1 ",10(2X,E10,4))	FUNCT174
103 FORMAT(" D2 ",10(2X,E10,4))	FUNCT175
104 FORMAT(" D3 ",10(2X,E10,4))	FUNCT176
105 FORMAT(" D4 ",10(2X,E10,4))	FUNCT177
106 FORMAT(" D5 ",10(2X,E10,4))	FUNCT178
112 FORMAT(" D6 ",10(2X,E10,4))	FUNCT179
107 FORMAT(" KPHI ",E10,4," RHOG ",E10,4," PHIH ",E10,4)	FUNCT180
109 FORMAT(" MASS ",E10,4," CINT ",E10,4," QA ",E10,4," CE ",E10,4,	FUNCT181
"CE2 ",E10,4," CE3 ",E10,4)	FUNCT182
110 FORMAT(" DMU ",E10,4," EDMU ",E10,4," E2DMU ",E10,4," E3DMU ",	FUNCT183
"E10,4," BF ",E10,4," BMM ",E10,4)	FUNCT184
113 FORMAT(4H Z1 ,10(2X,E10,4))	FUNCT185
114 FORMAT(4H Z2 ,10(2X,E10,4))	FUNCT186
115 FORMAT(4H Z3 ,10(2X,E10,4))	FUNCT187
116 FORMAT(4H Z4 ,10(2X,E10,4))	FUNCT188
118 FORMAT(4H Z5 ,10(2X,E10,4))	FUNCT189
119 FORMAT(4H Z6 ,10(2X,E10,4))	FUNCT190
120 FORMAT(4H Z7 ,10(2X,E10,4))	FUNCT191
117 FORMAT(5H ZMA ,E10,4,6H ZWMA ,E10,4,6H EMAS ,E10,4,	FUNCT192
7H ZZWMA ,E10,4,7H ZWEMA ,E10,4,7H ZZWMA ,E10,4,	FUNCT193
7H E2MAZ ,E10,4)	FUNCT194

111	CONTINUE	FUNCT195
	RETURN	FUNCT196
	END	FUNCT197
	SUBROUTINE COMPUT(X)	COMPUT 2
	DIMENSION X(6)	COMPUT 3
	REAL KAR,KPI	COMPUT 4
	REAL NL,MASS,NCG,M,IT,IA,K,MA,MMAX,N	COMPUT 5
	INTEGER END	COMPUT 6
C		COMPUT 7
	COMMON /SHIP/ MASS,CINT,QA,CE,CE2,CE3,DMU,EDMU,E2DMU,E3DMU,BF,BMM,	COMPUT 8
	NL,FL,IA,E(120)	COMPUT 9
	COMMON /CONST/ NCG,ECG,PI,DPR,RPD,GRAVITY,RHO,K,NUM,MA(120),CD,TA,	COMPUT10
	B(120),BETA,HW(120),TZ,DRAG,W,XD,T,XP,M,IT,	COMPUT11
	DELTA S,TX,EST(120),C,RO,KAR,MMAX(1 0),TEST(120),	COMPUT12
	N(120),PHALF	COMPUT13
	COMMON/OUT/NPRINT,NPLOT,END	COMPUT14
	COMMON /TEPMS/ T1,T2,T3,T4,T5,T6,T7,T8	COMPUT15
	COMMON /WAVE/ R(120),PT(120),ZMA,ZWMA,EMAS,ZZWMA,ZWEMA,ZZWMA,	COMPUT16
	E2MAZ,ZWDOT(120)	COMPUT17
	COMMON /TEST/ VMA	COMPUT18
C		COMPUT19
C		COMPUT20
	CX6 = COS(X(6))	COMPUT21
	SX6 = SIN(X(6))	COMPUT22
	W0 = K*C	COMPUT23
	PIH = PI/2.0	COMPUT24
	KPI = KAR*PI	COMPUT25
	CONS1 = RO*W0*W0*CX6	COMPUT26
	CONS2 = (KPI*RHO*PIH/TA)/CX6	COMPUT27
	CONS3 = RO*W0*K*CX6*SX6	COMPUT28
	CONS4 = RO*W0*K*CX6*CX6	COMPUT29
	TERM1 = X(1)*CX6	COMPUT30
	TERM2 = X(2)*SX6	COMPUT31
	UVNUM = (X(1)*CX6-(X(2)-ZWDOT(NUM))*SX6)*	COMPUT32
	(X(1)*SX6-X(3)*E(NUM)+(X(2)-ZWDOT(NUM))*CX6)	COMPUT33
C		COMPUT34
	ZMA = ZMA*X(3)*SX6	COMPUT35
	ZZWMA = ZZWMA*X(3)*SX6	COMPUT36
	ZWMA = ZWMA*CONS1	COMPUT37
	EMAS = EMAS*CONS1	COMPUT38
	DMU = DMU*CONS2	COMPUT39
	EDMU = EDMU*CONS2	COMPUT40
	CE = CE*CD*RHO	COMPUT41
	CE2 = CE2*CD*RHO	COMPUT42
	E2DMU = E2DMU*CONS3	COMPUT43
	E3DMU = E3DMU*CONS3	COMPUT44
	ZWEMA = ZWEMA*CONS4	COMPUT45
	ZZWMA = ZZWMA*CONS4	COMPUT46
C		COMPUT47
20	T1 = QA*X(3)*(TERM1-TERM2)	COMPUT48
	T1 = T1 + ZZWMA - EMAS	COMPUT49
	T2 = EDMU	COMPUT50
	T3 = CE2	COMPUT51
	T4 = MA(NUM)*E(NUM)*UVNUM + E2MAZ + E3DMU - ZZWMA + BMM	COMPUT52
	NL = T1 + T2 + T3 + T4 + BMM	COMPUT53
	T5 = MASS*X(3)*(TERM2-TERM1)	COMPUT54
	T5 = T5 + ZWMA - ZMA	COMPUT55
	T6 = -DMU	COMPUT56
	T7 = -CE	COMPUT57

T8 = -MA(NIJM)*UVNUM - E2DMU + ZWEMA	COMPUT58
BF = BF/CX6	COMPUT59
FL=T5+T6+T7+T8-BF	COMPUT60
IF(NPRINT,LT,3)GO TO 30	COMPUT61
25 CONTINUE	COMPUT62
WRITE(6,10)NL,FL	COMPUT63
10 FORMAT(" NL = ",E12.6," FL = ",E12.6)	COMPUT64
30 RETURN	COMPUT65
END	COMPUT66
SUBROUTINE INPUT	COMPUT67
C* * * * * DEFINITION OF INPUT VARIABLES	COMPUT68
C XA = INITIAL TIME	INPUT 2
C XE = FINAL TIME	INPUT 3
C HMIN = MINIMUM STEP SIZE	INPUT 4
C HMAX = MAXIMUM STEP SIZE	INPUT 5
C EPSE = RELATIVE ERROR CRITERIUM USED FOR VALUES OF Y GT A	INPUT 6
C EPS = ERROR CRITERION IN KUTMER	INPUT 7
C A = ABSOLUTE ERROR CRITERIA USED IN KUTMER	INPUT 8
C NPRINT = 1 FINAL PRINTOUT	INPUT 9
C = 2 MATRIX INVERSE MATRIX,F COLUMN MATRIX,AND KUTMER	INPUT 10
C RESULTS	INPUT 11
C = 3 INTEGRAL VALUES	INPUT 12
C = 4 CALCULATED VALUES-CONSTANT FOR GIVEN INPUT VALUES	INPUT 13
C NPLOT = 0 NO PLOT	INPUT 14
C = 1 PRINTER PLOT	INPUT 15
C END = NUMBER OF RUNS	INPUT 16
C	INPUT 17
C M = MASS OF CRAFT	INPUT 18
C W = WEIGHT OF CRAFT	INPUT 19
C TZ = THRUST COMPONENT IN Z DIRECTION	INPUT 20
C TX = THRUST COMPONENT IN X DIRECTION	INPUT 21
C XECG = DISTANCE FROM CG TO CENTER OF PRESSURE FOR NORMAL FORCE	INPUT 22
C XP = MOMENT ARM OF PROPELLER THRUST	INPUT 23
C XD = DISTANCE FROM CG TO CENTER OF PRESSURE FOR DRAG FORCE	INPUT 24
C KA(I)= ADDED MASS COEFFICIENT	INPUT 25
C AN ARRAY GIVEN THE VALUE KAR WHICH IS READ IN	INPUT 26
C BM(I) = BEAM AT FREE SURFACE OR AT CHINE	INPUT 27
C DRAG = FRICTION DRAG	INPUT 28
C K = WAVE NUMBER	INPUT 29
C RO = WAVE HEIGHT	INPUT 30
C NU = WAVE SLOPE	INPUT 31
C NUM = NUMBER OF STATIONS	INPUT 32
C BL = BOAT LENGTH	INPUT 33
C LAMBDA = WAVE LENGTH	INPUT 34
C RG = RADIUS OF GENERATION IN FEET	INPUT 35
C T = PROPELLED THRUST IN LBS	INPUT 36
C GAMMA = PROPELLER THRUST ANGLE IN DEGREES	INPUT 37
C DELTAS=STATION SPACING IN FEET	INPUT 38
C ECG = LONGITUDINAL CENTER OF GRAVITY	INPUT 39
C NCG = VERTICAL CG	INPUT 40
C BETA(I) = DEAD RISE	INPUT 41
C NO(I) = HEIGHT OF MEAN BUTTOCK	INPUT 42
C RHO = DENSITY OF WATER	INPUT 43
C GRAVITY = GRAVITY FT/SEC**2	INPUT 44
C DPR = DEGREES PER RADIAN	INPUT 45
C RPD = RADIAN PER DEGREE	INPUT 46
C PI = 3.14159	INPUT 47
	INPUT 48
	INPUT 49

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C EST(I) = STATION POSITION INPUT 50
C START = START TIME OF THE RAMP FUNCTION FOR SEA WAVE INPUT 51
C RISE = DURATION OF THE RISE FROM ZERO TO ONE OF THE RAMP INPUT 52
C INPUT 53
C * * * * * IC OPTIONS INPUT 54
C INPUT 55
C IC(1) =1 USE WAVE Z DISTANCE IN COMPUTING LIFT COMPONENT INPUT 56
C OF NL AND FL INPUT 57
C INPUT 58
C INPUT 59
C REAL IT,K,LAMBDA,M,MA,HMAX,NU,N,NCG,NO,MASS,NL,IA,KAR INPUT 60
C INTEGER END INPUT 61
C INPUT 62
C INPUT 63
C COMMON /CONST/ NCG,ECG,PI,DPR,RPD,GRAVITY,RHO,K,NUM,MA(120),CD,TA, INPUT 64
C B(120),BETA,HW(120),TZ,DRAG,W,XD,T,XP,M,IT, INPUT 65
C DELTAS,TX,EST(120),C,RO,KAR,HMAX(120),TEST(120), INPUT 66
C N(120),PHALF INPUT 67
C COMMON /SHIP/ MASS,CINT,QA,CE,CE2,CE3,DMU,EDMU,E2DMU,E3DMU,BF,BMM, INPUT 68
C NL,FL,IA,E(120) INPUT 69
C COMMON /IN/ BM(120),B1(120),VELIN INPUT 70
C COMMON /IN2/ NO(120),XA,XE,HMAX,HMIN,A(6),EPSE(6),LAMBDA INPUT 71
C COMMON/OUT/NPRINT,NPLOT,END INPUT 72
C COMMON /ACCEL/ XACCL,BWACL,CGACL,BL INPUT 73
C INPUT 74
C NAMELIST/HSP/A,NPRINT,NPLOT,END,W,HL,TZ,TX,XECG,XP,XD, INPUT 75
C DRAG,RG,T,GAMMA,ECG,NCG,KAR,RO,LAMBDA,NUM,BETA,EST INPUT 76
C ,XA,XE,HMIN,HMAX,EPS,VELIN INPUT 77
C INPUT 78
C DATA A /.01, .0001, .00001, .1, .0001, .00001/ INPUT 79
C DATA NPRINT,NPLOT,END/.1,1,1/ INPUT 80
C DATA W,BL,TZ,TX,XECG,XP,XD,DRAG,RU,LAMBDA,RG,T,GAMMA, INPUT 81
C ECG,NCG,KAR /16, .3.75,6*0.0, .0416,22.5, .9562,2*0.0, INPUT 82
C 2.325,0.0,1.0/ INPUT 83
C DATA NUM,BETA,EST /77,20.0, INPUT 84
C 0.0000, .03125, .06250, .09375, .12500, .15625, .18750, .21875, INPUT 85
C .25000, .28125, .31250, .34375, .37500, .40625, .43750, .46875, INPUT 86
C .50000, .53125, .56250, .59375, .62500, .65625, .6875, .71875, INPUT 87
C .75000, .78125, .81250, .84375, .87500, .90625, .93750, .96875,1.000, INPUT 88
C 1.06250,1.12500,1.18750,1.25000,1.3125,1.37500,1.4375, INPUT 89
C 1.500,1.5625,1.625,1.6875,1.75,1.8125,1.875,1.9375,2.0, INPUT 90
C 2.0625,2.125,2.1875,2.25,2.3125,2.375,2.4375,2.5,2.5625,2.625, INPUT 91
C 2.6875,2.75,2.8125,2.8750,2.9375,3.0,3.0625,3.125,3.1875, INPUT 92
C 3.2500,3.3125,3.375,3.4375,3.5,3.5625,3.625,3.6875,3.75 / INPUT 93
C DATA XA,XE,HMIN,HMAX,EPS /0.0,20.0, .025, .1, .15/ INPUT 94
C DATA VELIN /19.62/ INPUT 95
C INPUT 96
C * * * * * READ IN AND WRITE OUT KUTMER PARAMETERS AND PROGRAM INPUT 97
C OPTIONS INPUT 98
C READ(5,HSP) INPUT 99
C WRITE(6,HSP) INPUT100
C DO 10 I=1,6 INPUT101
C 10 EPSE(I) = EPS INPUT102
C INPUT103
C * * * * * SET UP CONSTANTS INPUT104
C PI = 3.141592653589 INPUT105
C GRAVITY=32.18 INPUT106
C DPR=57.29577951308 INPUT107
C RPD=.017453292519 INPUT108

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IF(EST(NUM),LT,3.75) STOP 3	INPUT109
C	INPUT110
COMPUTE NO AND RM ARRAYS	INPUT111
C	INPUT112
DO 32 I=1,NUM	INPUT113
IF(EST(I),GE,0.75) GO TO 30	INPUT114
NO(I)=-0.46875*(1.0-SQRT(EST(I)/0.375-(EST(I)/0.75)**2.0))	INPUT115
BM(I)=.375*SQRT(1.0-(EST(I)/.75-1.0)**2.0)	INPUT116
GO TO 32	INPUT117
30 NO(I)=0.0	INPUT118
BM(I) = 0.375	INPUT119
32 CONTINUE	INPUT120
C*****COMPUTE CONSTANTS AND INITIALIZE ARRAYS	INPUT121
M=W/GRAVITY	INPUT122
RHO=1.99	INPUT123
IT=M*RG*RG	INPUT124
K = 2.*PI/LAMBDA	INPUT125
C=SQRT(GRAVITY/K)	INPUT126
NU=RO*K	INPUT127
PHALF = (PI/2.)*RHO	INPUT128
C	INPUT129
BETA = BETA*RPD	INPUT130
CD = COS(BETA)	INPUT131
TA = TAN(BETA)	INPUT132
DO 60 I=1,NUM	INPUT133
E(I) = ECG-EST(I)	INPUT134
N(I) = NCG-NO(I)	INPUT135
MMA(I) = KAR*PHALF*BM(I)*BM(I)	INPUT136
TEST(I) = (2.*BM(I)*TA)/PI	INPUT137
60 CONTINUE	INPUT138
END=END+1	INPUT139
RETURN	INPUT140
END	INPUT141
SUBROUTINE PLOTTER(FX,XA,HMAX,LAMBDA,IB,NWAVE,IPT)	PLOTTER 2
C	PLOTTER 3
INPUT:	PLOTTER 4
C FX A TWO DIMENSIONAL ARRAY CONTAINING PITCH AND	PLOTTER 5
C HEAVE VALUES AT EACH TIME STEP	PLOTTER 6
C XA INITIAL TIME	PLOTTER 7
C HMAX TIME INTERVAL, PTIME*HMAX = INTERVAL BETWEEN	PLOTTER 8
C FX VALUES	PLOTTER 9
C LAMBDA WAVELENGTH USED IN CALCULATING PITCH AND	PLOTTER10
C HEAVE RATIOS	PLOTTER11
C IB NUMBER OF FX VALUES	PLOTTER12
C NWAVE START OF VALUES AFTER WAVE IS COMPLETELY ON	PLOTTER13
C	PLOTTER14
C	PLOTTER15
REAL IT,K,LAMBDA,M,MA,MMA,N,NCG	PLOTTER16
INTEGER END	PLOTTER17
C	PLOTTER18
DIMENSION FX(2,400),FMIN(2),FMAX(2),NVAR(2)	PLOTTER19
C	PLOTTER20
COMMON /CONST/ NCG,ECG,PI,DPR,RPD,GRAVITY,RHO,K,NUM,MA(120),CD,TA,	PLOTTER21
• B(120),BETA,HW(120),TZ,DRAG,W,XD,T,XP,M,IT,	PLOTTER22
• DELTAS,TX,EST(120),C,RO,KA,MMA(120),TEST(120),	PLOTTER23
• N(120),PHALF	PLOTTER24
COMMON/OUT/NPRINT,NPLOT,END	PLOTTER25
C	PLOTTER26
C * * * * * SET UP VALUES FOR PLOT AND CREATE PLOT	PLOTTER27

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      NFUN=2
C * * * * * SET UP MIN AND MAX LIMITS FOR PLOT
      FMIN(1)=FX(1,1)
      FMIN(2)=FX(2,1)
      FMAX(1)=FX(1,1)
      FMAX(2)=FX(2,1)
C * * * * * SET UP MIN AND MAX LIMITS FOR PITCH AND HEAVE RATIO
      FMNP=FX(2,NWAVE)
      FMXP=FX(2,NWAVE)
      FMNH=FX(1,NWAVE)
      FMXH=FX(1,NWAVE)
C
      DO 200 I=1,IB
      IF (FX(1,I).LT.FMIN(1)) FMIN(1)=FX(1,I)
      IF (FX(1,I).GT.FMAX(1)) FMAX(1)=FX(1,I)
      IF (FX(2,I).LT.FMIN(2)) FMIN(2)=FX(2,I)
      IF (FX(2,I).GT.FMAX(2)) FMAX(2)=FX(2,I)
      IF (1.LE.NWAVE) GO TO 200
      IF (FX(1,I).LT.FMNH) FMNH=FX(1,I)
      IF (FX(1,I).GT.FMXH) FMXH=FX(1,I)
      IF (FX(2,I).LT.FMNP) FMNP=FX(2,I)
      IF (FX(2,I).GT.FMXP) FMXP=FX(2,I)
200 CONTINUE
      IF (IPT.EQ.0) GO TO 800
C * * * * * COMPUTE RATIOS
      COL3 = (FMXH-FMNH)/(2.*RO)
      COL4 = (FMXP-FMNP)/((4.*PI*RO)/LAMBDA)
      WRITE(4,700) COL3,COL4
700 FORMAT(1H1," HEAVE AMPLITUDE/WAVEHEIGHT = ",E12.6,/,2X,
. " PITCH AMPLITUDE/(2.*PI*WAVEHEIGHT/LAMBDA) = ",E12.6)
C
800 CONTINUE
      NVAR(1)=10H HEAVE
      NVAR(2)=10H PITCH
      N1=2
      X0=XA
      DELX = HMAX
      IF (NPLOT.EQ.1) CALL PLOT2(FX,FMIN,FMAX,NVAR,NFUN,N1,IB,X0,DELX)
      RETURN
      END
      SUBROUTINE TRAP(F,DX,NPTS,ANS)
C
C INPUT:
C      F          ARRAY OF FUNCTIONAL VALUES OF THE INTEGRAND
C      DX         THE X INTERVAL BETWEEN VALUES
C      NPTS       THE NUMBER OF VALUES GIVEN
C OUTPUT:
C      ANS        THE VALUE OF THE INTEGRAL
C
      DIMENSION F(NPTS)
      ANS=0.0
      IF (NPTS.LT.2) GO TO 999
      DO 1 I=1,NPTS
1      ANS=ANS+F(I)
      ANS=DX*(ANS-0.5*(F(1)+F(NPTS)))
999 CONTINUE
      RETURN
      END
      FUNCTION RMP(T,START,RISE)

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PLOTERR28
PLOTERR29
PLOTERR30
PLOTERR31
PLOTERR32
PLOTERR33
PLOTERR34
PLOTERR35
PLOTERR36
PLOTERR37
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PLOTERR54
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PLOTERR58
PLOTERR59
PLOTERR60
PLOTERR61
PLOTERR62
PLOTERR63
PLOTERR64
PLOTERR65
PLOTERR66
PLOTERR67
TRAP 2
TRAP 3
TRAP 4
TRAP 5
TRAP 6
TRAP 7
TRAP 8
TRAP 9
TRAP 10
TRAP 11
TRAP 12
TRAP 13
TRAP 14
TRAP 15
TRAP 16
TRAP 17
TRAP 18
TRAP 19
RMP 2

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C	***** THIS FUNCTION IS USED TO GRADUALLY IMPLIMENT THE WAVE	RMP	3
C		RMP	4
C	T CURRENT TIME	RMP	5
C	START TIME TO START RAMP FROM 0.0 TO 1.0	RMP	6
C	RISE THE LENGTH OF THE RISE FROM 0.0 TO 1.0	RMP	7
C		RMP	8
	H=0.0	RMP	9
	IF(T.LT.START)GO TO 99	RMP	10
	IF(RISE.EQ.0.0)GO TO 80	RMP	11
	TOP=T-START	RMP	12
	H=1.0	RMP	13
	IF(TOP.LT.RISE)H=TOP/RISE	RMP	14
	GO TO 99	RMP	15
80	H=1.	RMP	16
	IF(T.EQ.START)H=0.5	RMP	17
99	RMP=H	RMP	18
	RETURN	RMP	19
	END	RMP	20

LISTING OF COMPUTER PROGRAM FOR CALCOMP PLOTS

	PROGRAM PLTHSP(INPUT,OUTPUT,TAPE5=INPUT,TAPE6=OUTPUT,TAPE7,TAPE9)	MAIN	2
	ITAPE = 7	MAIN	3
	CALL CALPLT(ITAPE)	MAIN	4
	STOP	MAIN	5
	END	MAIN	6
	SUBROUTINE CALPLT(ITAPE)	CALP	2
	DIMENSION TIME(4003),PITCH(4003),HEAVE(4003)	CALP	3
	*,IBUF(1000),BWACL(4003),CGACL(4003)	CALP	4
	LOGICAL ACCEL	CALP	5
C		CALP	6
C	CAL CUMP PLOT OF PITCH AND HEAVE VERSUS TIME	CALP	7
C		CALP	8
	IREAD = 5	CALP	9
	READ(IREAD,10) XAXIS,YAXISP,YAXISH,HT	CALP	10
10	FORMAT(8F10.0)	CALP	11
	ACCEL = .FALSE.	CALP	12
	READ(IREAD,20) IA	CALP	13
20	FORMAT(110)	CALP	14
	IF(IA.EQ.1) ACCEL = .TRUE.	CALP	15
	IF(ACCEL) READ(IREAD,10) YAXISB,YAXISC	CALP	16
	CALL READT(TIME,HEAVE,PITCH,BWACL,CGACL,NPTS)	CALP	17
	CALL PLOTS(IBUF,1000,7)	CALP	18
	CALL PLOT(0.5,1.0,-3)	CALP	19
	CALL ESCALE(TIME,XAXIS,NPTS,1)	CALP	20
	CALL ESCALE(HEAVE,YAXISH,NPTS,1)	CALP	21
	CALL ESCALE(PITCH,YAXISP,NPTS,1)	CALP	22
	IF(ACCEL) CALL ESCALE(BWACL,YAXISB,NPTS,1)	CALP	23
	IF(ACCEL) CALL ESCALE(CGACL,YAXISC,NPTS,1)	CALP	24
	N1 = NPTS+1	CALP	25
	N2 = NPTS+2	CALP	26
	N3 = NPTS+3	CALP	27
	CALL EAXIS(0.0,0.0,15HTIME IN SECONDS,-15,XAXIS,0.0,	CALP	28
	TIME(N1),TIME(N2),TIME(N3),HT)	CALP	29
	CALL EAXIS(0.0,0.0,13HEAVE IN FEET,13,YAXISH,90.0,	CALP	30
	HEAVE(N1),HEAVE(N2),HEAVE(N3),HT)	CALP	31
	TEMP = TIME(N2)	CALP	32
	TIME(N2) = TIME(N2)/TIME(N3)	CALP	33
	HEAVE(N2) = HEAVE(N2)/HEAVE(N3)	CALP	34
	CALL LINE(TIME,HEAVE,NPTS,1,0,0)	CALP	35
	TIME(N2) = TEMP	CALP	36
	XNEW = XAXIS*3.	CALP	37
	YNEW = 1.0	CALP	38
	CALL PLOT(XNEW,0.0,-3)	CALP	39
	CALL EAXIS(0.0,0.0,15HTIME IN SECONDS,-15,XAXIS,0.0,	CALP	40
	TIME(N1),TIME(N2),TIME(N3),HT)	CALP	41
	CALL EAXIS(0.0,0.0,13HPITCH IN RAU.,13,YAXISP,90.0,	CALP	42
	PITCH(N1),PITCH(N2),PITCH(N3),HT)	CALP	43
	TIME(N2) = TIME(N2)/TIME(N3)	CALP	44
	PITCH(N2) = PITCH(N2)/PITCH(N3)	CALP	45
	CALL LINE(TIME,PITCH,NPTS,1,0,0)	CALP	46
	IF(.NOT.ACCEL) GO TO 30	CALP	47
	TIME(N2) = TEMP	CALP	48
	CALL PLOT(XNEW,0.0,-3)	CALP	49
	CALL EAXIS(0.0,0.0,15HTIME IN SECONDS,-15,XAXIS,0.0,TIME(N1),	CALP	50
	TIME(N2),TIME(N3),HT)	CALP	51
	CALL EAXIS(0.0,0.0,16HRAW ACCELERATION,16,YAXISB,90.0,BWACL(N1),	CALP	52
	BWACL(N2),BWACL(N3),HT)	CALP	53
	TIME(N2) = TIME(N2)/TIME(N3)	CALP	54
	BWACL(N2) = BWACL(N2)/BWACL(N3)	CALP	55

	CALL LINE (TIME,BWACL,NPTS,1,0,0)	CALP	56
C	TIME(N2) = TEMP	CALP	57
	CALL PLOT(XNEW,0,0,-3)	CALP	58
	CALL EAXIS(0,0,0,0,15HTIME IN SECONDS,-15,XAXIS,0,0,TIME(N1),	CALP	59
	TIME(N2),TIME(N3),HT)	CALP	60
*	CALL EAXIS(0,0,0,0,15HCG ACCELERATION,15,YAXIS,90,0,CGACL(N1),	CALP	61
	CGACL(N2),CGACL(N3),HT)	CALP	62
*	TIME(N2) = TIME(N2)/TIME(N3)	CALP	63
	CGACL(N2) = CGACL(N2)/CGACL(N3)	CALP	64
	CALL LINE (TIME,CGACL,NPTS,1,0,0)	CALP	65
30	CONTINUE	CALP	66
	CALL PLOT(10,0,0,0,999)	CALP	67
	RETURN	CALP	68
	END	CALP	69
	SUBROUTINE READT (TIME,HEAVE,PITCH,BWACL,CGACL,NPTS)	CALP	70
	DIMENSION X(6),HEAVE(1),PITCH(1)	READ	2
	TIME(1),BWACL(1),CGACL(1)	READ	3
	I = 0	READ	4
5	CONTINUE	READ	5
	I = I+1	READ	6
	READ(9) TIME(I),(X(I),I=4,6),BWACL(I),CGACL(I)	READ	7
	IF (EOF(9)) 10,15	READ	8
15	CONTINUE	READ	9
	WRITE(6,20) TIME(I),(X(J),J=4,6),BWACL(I),CGACL(I)	READ	10
20	FORMAT(1H ,6(F7.2,2X))	READ	11
	HEAVE(I) = X(5)	READ	12
	PITCH(I) = X(6)	READ	13
	IF (I,GE,4000) GO TO 10	READ	14
	GO TO 5	READ	15
10	CONTINUE	READ	16
	NPTS = 1-1	READ	17
	RETURN	READ	18
	END	READ	19
	SUBROUTINE EAXIS(XPAGE,YPAGE,IBCD,NCHAR,AXLEN,ANGLE,FIRSTV,	READ	20
	DELTAU,DELTAU,HT)	EAXIS	2
	DIMENSION IBCD(1)	EAXIS	3
C		EAXIS	4
C	THIS ROUTINE WORKS LIKE THE CALCUMP AXIS WITH THE	EAXIS	5
C	EXCEPTION THAT THE TICK MARKS ARE NOT NECESSARILY	EAXIS	6
C	EVERY INCH AND THE HEIGHT OF THE CHARACTERS IS INPUTTED	EAXIS	7
C		EAXIS	8
	CALL PLOT(XPAGE,YPAGE,3)	EAXIS	9
	ISN = ISIGN(1,NCHAR)	EAXIS	10
	ISGN = SIGN(1.,DELTAU)	EAXIS	11
	AMIN = FIRSTV	EAXIS	12
	X = XPAGE	EAXIS	13
	Y = YPAGE	EAXIS	14
	XNUM = FIRSTV-DELTAU	EAXIS	15
	N = AXLEN/DELTAU	EAXIS	16
	IF (N*DELTAU.LT,AXLEN) N=N+1	EAXIS	17
	AMAX = AMIN+(N*DELTAU)	EAXIS	18
	NDIG = NDIGIT(AMIN,AMAX,DELTAU,ND)	EAXIS	19
10	CONTINUE	EAXIS	20
	TEST = (NDIG*HT) + HT	EAXIS	21
	IF (TEST.GT,DELTAU) HT=HT/2.	EAXIS	22
	IF (TEST.GT,DELTAU) GO TO 10	EAXIS	23
	AYN = (1.5*HT)	EAXIS	24
	BYN = ((NDIG-2)*HT)/2.+.5*HT)	EAXIS	25
		EAXIS	26

N = N+1	EAXIS 27
TANG = (90.+ANGLE)/57.2958	EAXIS 28
ANG = ANGLE/57.2958	EAXIS 29
ST = SIN(TANG)	EAXIS 30
CT = COS(TANG)	EAXIS 31
S = SIN(ANG)	EAXIS 32
C = COS(ANG)	EAXIS 33
DO 30 I=1,N	EAXIS 34
IF(I.EQ.1) GO TO 20	EAXIS 35
X = X+DELTAU*C	EAXIS 36
Y = Y+DELTAU*S	EAXIS 37
CALL PLOT(X,Y,2)	EAXIS 38
IF(I.EQ.N) GO TO 20	EAXIS 39
XT = X+(.1*CT*ISN)	EAXIS 40
YT = Y+(.1*ST*ISN)	EAXIS 41
CALL PLOT(XT,YT,2)	EAXIS 42
20 XN = X+AYN*CT*ISN-BYN*C	EAXIS 43
YN = Y+AYN*ST*ISN-BYN*S	EAXIS 44
XNUM = XNUM+DELTAV	EAXIS 45
CALL NUMBER(XN,YN,HT,XNUM,ANGLE,ND)	EAXIS 46
CALL PLOT(X,Y,3)	EAXIS 47
30 CONTINUE	EAXIS 48
XSP = (((AXLEN/HT)/2.)-(IABS(NCHAR)/2.))*HT	EAXIS 49
YSP = 3.5*HT	EAXIS 50
XT = XPAGE + XSP*C + ISN*YSP*CT	EAXIS 51
YT = YPAGE + XSP*S + ISN*YSP*ST	EAXIS 52
CALL SYMBOL(XT,YT,HT,IBCD,ANGLE,IABS(NCHAR))	EAXIS 53
RETURN	EAXIS 54
END	EAXIS 55
FUNCTION NDIGIT(AMIN,AMAX,ANUM,ND)	NDIG 2
C	NDIG 3
C FINDS THE NUMBER OF DIGITS NECESSARY TO PRINT	NDIG 4
C EVEN INCREMENT OF THE FUNCTION ON THE AXIS	NDIG 5
C	NDIG 6
C NDIGIT THE NUMBER OF PLACES IN THE ENTIRE NUMBER	NDIG 7
C ND THE NUMBER OF DECIMAL PLACES	NDIG 8
C ANUM THE VALUE GIVEN TO EACH INCREMENT ON THE AXIS	NDIG 9
C	NDIG 10
IF(ABS(AMIN).LT.ABS(AMAX)) GO TO 20	NDIG 11
IF(ABS(AMIN).EQ.ABS(AMAX).AND.AMAX.NE.0) GO TO 20	NDIG 12
IF(ABS(AMIN).GT.ABS(AMAX)) GO TO 10	NDIG 13
AMAX = 1.	NDIG 14
AMIN = -1.	NDIG 15
GO TO 20	NDIG 16
10 AMAX = ABS(AMIN)	NDIG 17
20 IF(AMAX.LE.1.) GO TO 50	NDIG 18
NDIV = 10	NDIG 19
I = 1	NDIG 20
30 IF(AMAX/NDIV.LT.1) GO TO 40	NDIG 21
I = I+1	NDIG 22
NDIV = NDIV*10	NDIG 23
GO TO 30	NDIG 24
40 NDIGIT = I+3	NDIG 25
ND = 2	NDIG 26
GO TO 80	NDIG 27
50 NDIV = 10	NDIG 28
I = 1	NDIG 29
60 IF(AMAX*NDIV.GT.1.) GO TO 70	NDIG 30
I = I+1	NDIG 31

	NDIV = NDIV*10	NDIG 32
	GO TO 60	NDIG 33
70	NDIGIT = I+2	NDIG 34
	ND = I	NDIG 35
80	DD = FLUAT(ND)	NDIG 36
	X = ANUM*(10**DD)	NDIG 37
	IX = X	NDIG 38
	IF(X-FLUAT(IX).LT..0001) GO TO 90	NDIG 39
	DD = DD+1	NDIG 40
	ND = ND+1	NDIG 41
	NDIGIT = NDIGIT+1	NDIG 42
	GO TO 80	NDIG 43
90	CONTINUE	NDIG 44
	RETURN	NDIG 45
	END	NDIG 46
	SUBROUTINE ESCALE (ARRAY,AXLEN,NPTS,INC)	ESCAL 2
C		ESCAL 3
C	FINDS THE SCALE TO BE USED ON THE AXIS -	ESCAL 4
C	ARRAY MUST HAVE THREE UNUSED POSITIONS	ESCAL 5
C	ARRAY(NPTS+1) = FIRSTV	ESCAL 6
C	ARRAY(NPTS+2) = DELTAV (THE INCREMENT BETWEEN TICK MARKS	ESCAL 7
C	VALUES - NUMBERS)	ESCAL 8
C	ARRAY(NPTS+3) = DELTAU (THE INCREMENT IN INCHES	ESCAL 9
C	BETWEEN TICK MARKS)	ESCAL 10
C		ESCAL 11
C		ESCAL 12
C	DIMENSION ARRAY(1)	ESCAL 13
	AMIN = ARRAY(1)	ESCAL 14
	AMAX = ARRAY(1)	ESCAL 15
	ISGN = ISIGN(1,INC)	ESCAL 16
	INC = IABS(INC)	ESCAL 17
	DO 10 I=1,NPTS,INC	ESCAL 18
	IF (ARRAY(I).LT.AMIN) AMIN=ARRAY(I)	ESCAL 19
	IF (ARRAY(I).GT.AMAX) AMAX=ARRAY(I)	ESCAL 20
10	CONTINUE	ESCAL 21
20	AUNIT = IUNIT (AMIN,AMAX,AXLEN,N,ANUM)	ESCAL 22
	CALL ADJUST (AMIN,AMAX,AUNIT,AXLEN,N,ANUM)	ESCAL 23
	ARRAY(NPTS+1) = AMIN	ESCAL 24
	ARRAY(NPTS+2) = ANUM*ISGN	ESCAL 25
	IF (ISGN.EQ.-1) ARRAY(NPTS+1) = AMAX	ESCAL 26
	ARRAY(NPTS+3) = AUNIT	ESCAL 27
	IF (ABS(ANUM).EQ.AUNIT) ARRAY(NPTS+2) = 1.*ISGN	ESCAL 28
	IF (ABS(ANUM).EQ.AUNIT) ARRAY(NPTS+3) = 1.	ESCAL 29
	RETURN	ESCAL 30
	END	ESCAL 31
	SUBROUTINE ADJUST (AMIN,AMAX,AUNIT,AXLEN,N,ANUM)	JUST 2
C		JUST 3
C	GIVEN AMIN AND AMAX WHICH ARE DISTINCT VALUES, ADJUST	JUST 4
C	THEM SO THAT THEY ARE EVEN MULTIPLES OF AUNIT	JUST 5
C		JUST 6
	K = 1	JUST 7
	MIN = AMIN/ANUM	JUST 8
	IF (AMIN.LT.MIN*ANUM) MIN = MIN-1	JUST 9
	AMIN = MIN*ANUM	JUST 10
	MAX = AMAX/ANUM	JUST 11
	IF (AMAX.GT.MAX*ANUM) MAX = MAX+1	JUST 12
	AMAX = MAX*ANUM	JUST 13
10	TERM = AMIN+(N-K)*ANUM	JUST 14
	IF (TERM.LT.AMAX) GO TO 20	JUST 15

K = K+1	JUST	16
GO TO 10	JUST	17
20 AUNIT = AXLEN/(N-K+1)	JUST	18
N = AXLEN/AUNIT+1	JUST	19
RETURN	JUST	20
END	JUST	21
FUNCTION UNIT(AMIN,AMAX,AXLEN,N,ANUM)	UNIT	2
C	UNIT	3
C FINDS THE INCREMENT BETWEEN VALUES TO BE USED ON THE	UNIT	4
C AXIS IN AS FAR AS LABELING THE TICK MARKS	UNIT	5
C FINDS THE NUMBER OF DIVISIONS TO BE MADE ON THE AXIS	UNIT	6
C FINDS THE SIZE IN INCHES OF THESE DIVISIONS	UNIT	7
C	UNIT	8
IF(AMIN.NE.AMAX) GO TO 10	UNIT	9
AMIN = AMIN-1	UNIT	10
AMAX = AMAX+1	UNIT	11
10 IF(AMAX.LT.1.AND.AMIN.GT.-1)GO TO 110	UNIT	12
30 MIN = AMIN	UNIT	13
MAX = AMAX	UNIT	14
IF(AMAX.GT.MAX) MAX=MAX+1	UNIT	15
IF(AMIN.LT.MIN) MIN=MIN-1	UNIT	16
IF(MIN.LT.0) NWID = MAX+ABS(MIN)	UNIT	17
IF(MIN.GE.0) NWID = MAX-MIN	UNIT	18
NUM = 10	UNIT	19
40 IF(NWID.LT.NUM) GO TO 60	UNIT	20
NUM = NUM*10	UNIT	21
GO TO 40	UNIT	22
60 N = NWID/(NUM/10)	UNIT	23
IF(MIN.LT.0.AND.MAX.GT.0) GO TO 70	UNIT	24
IF(N*(NUM/10).LT.NWID) N=N+1	UNIT	25
ANUM = NUM/10.	UNIT	26
AUNIT = AXLEN/N	UNIT	27
GO TO 160	UNIT	28
70 NN = IABS(MIN)/(NUM/10)	UNIT	29
IF(NN*(NUM/10).LT.IABS(MIN)) NN = NN+1	UNIT	30
N = MAX/(NUM/10)	UNIT	31
IF(N*(NUM/10).LT.MAX) N = N+1	UNIT	32
N = N+NN	UNIT	33
ANUM = NUM/10.	UNIT	34
AUNIT = AXLEN/N	UNIT	35
GO TO 160	UNIT	36
110 NUM=10	UNIT	37
120 IF(AMAX*NUM.GT.1) GO TO 130	UNIT	38
NUM = NUM*10	UNIT	39
GO TO 120	UNIT	40
130 UNITT = 1./NUM	UNIT	41
140 N1 = AMIN*NUM	UNIT	42
N2 = AMAX*NUM	UNIT	43
IF(AMIN*NUM.LT.N1) N1=N1-1	UNIT	44
IF(AMAX*NUM.GT.N2) N2=N2+1	UNIT	45
IF(N1.NE.N2) GO TO 150	UNIT	46
AMIN = AMIN-UNITT	UNIT	47
AMAX = AMAX-UNITT	UNIT	48
GO TO 140	UNIT	49
150 N = N2-N1	UNIT	50
ANUM = UNITT	UNIT	51
IF(AMIN.LT.0.AND.AMAX.LT.0) N=N1-N2	UNIT	52
IF(AMIN.LT.0.AND.AMAX.GE.0) N=N2-N1	UNIT	53
AUNIT = AXLEN/N	UNIT	54

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160  IF(N.GT.5) GO TO 170
      N = N*2
      ANUM = ANUM/2.
      AUNIT = AUNIT/2.
      GO TO 160
170  UNIT = AUNIT
      RETURN
      END

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UNIT 55
UNIT 56
UNIT 57
UNIT 58
UNIT 59
UNIT 60
UNIT 61
UNIT 62

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